Perceived Income Risk Across the Income Distribution*

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Abstract

In incomplete-market models, the risk an agent perceives is the risk an econometrician estimates from realised earnings. Using data on perceptions, I show that this assumption is violated systematically across the income distribution. Conditional on staying in the same job, workers at the bottom perceive about double the wage risk of those at the top, hold that belief more persistently, and revise it more strongly after a shock, all of which has no counterpart in the realised process I estimate from matched earnings spells. I fit a learning rule that reproduces all three artefacts, and find that perceived risk departs from the truth in two directions at once: the bottom reports transitory risk far above its realised counterpart, while the middle and the top under-perceive their permanent risk. I trace the gradient to how legibly pay separates into a permanent and a transitory part, which I proxy using the itemised pay components of payment records. Embedded in a life-cycle buffer-stock economy, the measured beliefs open a precautionary-saving and marginal-propensity-to-consume wedge that rises with income and concentrates at the top, where perceived permanent risk falls furthest below the truth.

1 Introduction

In an incomplete-market economy, how much risk a household believes it faces is a primitive of its consumption and saving behaviour. Holding expected income and preferences fixed, an agent who perceives more risk to her future earnings accumulates a larger precautionary buffer and spends less out of a windfall, due to for instance prudence in the utility function or an occasionally binding borrowing constraint that induces precautionary saving. The size and the heterogeneity of idiosyncratic income risk are accordingly among the central inputs of the heterogeneous-agent models now standard in macroeconomics, where they shape the distribution of wealth, the distribution of marginal propensities to consume, and, through both, the transmission of aggregate shocks and of policy. But is the risk these models assign their agents the same as agents actually perceive, and if not, does the gap between the two fall evenly across households?

The common practice in this literature is to calibrate income risk indirectly. The econometrician estimates the cross-sectional dispersion of realised earnings in panel data, decomposes it into components of differing persistence (Meghir and Pistaferri, 2004), and hands the resulting variances to the agent as the risk she perceives. Perceived risk is set equal to realised risk: the agent is assumed to know exactly the variances the econometrician recovers, and to know nothing the econometrician cannot. The assumption is convenient but strong, and direct data on perceptions

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now allow it to be tested.

Measured directly, perceived risk violates this assumption, and it does so systematically across the income distribution. In the New York Fed's Survey of Consumer Expectations (SCE), which elicits from each respondent a density forecast of her own wage growth over the next twelve months, workers in households earning under $50k, conditional on staying in the same job, (i) report wage uncertainty about 1.8 times as large as those in households above $100k, (ii) hold their beliefs more persistently within person, and (iii) revise them more strongly after a realised wage shock. None of the three appears in the realised wage process: matched same-job spells in the Survey of Income and Program Participation (SIPP) carry no comparable gradient, and if anything realised risk rises mildly with income. The gradient must therefore originate in beliefs, and the practice of equating perceived with realised risk fails on all three aforementioned margins. This paper documents these patterns, fits a learning rule that reproduces them, and traces them to how pay is delivered across the income distribution, in three steps.

In the first step, Section 2 sets out the measurement, and Sections 3 and 4 establish the belief gradient and its dynamics, which are both novel findings in the literature, against the realised process and so fix what the rule must reproduce.

The second step fits a belief-updating rule. Through the signal-extraction lens of Pischke (1995), I conjecture that the agent cannot separate her permanent and transitory income risk shocks; each period she sees a pay change and must learn the variance of her permanent earning power from it. In Section 5, I model her belief as an exponentially weighted average of past squared surprises, which is simply the steady-state Kalman filter for an unknown variance that follows a random walk. The rule has three interpretable parameters: how fast beliefs update, what share of each squared surprise the agent reads as news about permanent earning power, and a prior on transitory risk as a level scaler for baseline risk.

Section 6 estimates the rule and finds that the bottom updates most slowly, books about three times the share of each squared surprise as permanent that the top does, a decline roughly twice what full information would imply, and holds a prior almost four times its realised transitory risk, while the middle and the top sit close to one another, at or below their realised processes. The persistence and prior gradients are statistically significant in the baseline; the attribution gradient is monotone but not.

The third step, in Section 7, asks why the deviations take the form they do. The rule reproduces the gradient but does not explain it, since its three parameters are estimated freely group by group. Measured against the risk each group actually faces, the bottom over-perceives its transitory risk, while the middle and the top under-perceive their permanent risk. No one-sided friction can produce a cross-section that sits on both sides of the truth, so I look for two forces of opposite sign, and locate both in how pay is delivered to a household.

My proposed microfoundation focuses on how legibly a worker's pay separates into its permanent and transitory parts. At the top, transitory pay arrives labelled and against a single salary, so the worker nets it out before she updates and her perceived risk falls. At the bottom it arrives fragmented, as variable shifts, cash tips, and irregular supplements layered on uncertain hours, so she cannot separate the transitory part, and the unresolved reading inflates her reported risk instead. Because the SIPP itemises the components of a paycheck, I can proxy legibility on the same stayer panel that supplies the realised process by the labelled share of within-spell pay movement: it rises from about an eighth at the bottom to over a third at the top, and the transitory risk it leaves unlabelled predicts, from the pay records alone, the level at which the middle and top priors sit, to within a standard error. The same share sets the persistence of beliefs, their response to news, and the under-perception of permanent risk higher up, since where pay is legible the worker faces a sharper signal about her permanent part and under-reads it, holding her attribution fixed as netting cleans the signal. The income gradient in beliefs therefore is a gradient in how pay is delivered, instead of how risky realised pay is.

Before any quantitative simulation, I analytically derive what the estimated mechanism implies for an agent's behaviour. In Section 8 I show that there exists a gap between an agent's perceived second moment and the true one, which results in a perceived-risk wedge. Because it is an error in a variance, it reaches saving through prudence, where a level error would act through the intertemporal-substitution motive. It bears on the perceived risk to permanent earning power, and hardly at all on the transitory prior, which a buffer absorbs within the period.

In Sections 9 and 10 I embed the measured beliefs in a life-cycle buffer-stock economy in which workers drift across income groups as their permanent earning power accumulates. Beyond their beliefs, the groups differ only in the risk they face, the realised wage variances and the job-loss and job-finding rates, so that the cross-section of wealth and marginal propensities is an outcome of the risk process. Relative to an otherwise identical full-information calibration, the measured beliefs open a precautionary-saving and marginal-propensity-to-consume (MPC) wedge. The saving wedge rises with income, almost three times as large at the top as at the bottom under the measured shares and roughly twice that size when the estimated belief responses are taken at face value, while the wedge in the MPC concentrates at the top, whose permanent belief is discounted most, liquidity reinforcing the ordering only at the margin. The bottom holds the most distorted transitory prior of any group, yet its permanent belief is close to the truth, so its behavioural wedge is small. In one-asset general equilibrium the belief economy supplies modestly less capital at a slightly higher interest rate, so the aggregates move less than the cross-section.

Section 11 traces the same beliefs through a legibility reform, a recession, and a risk shock; the reform is the mechanism's own experiment, raising the labelled share of low-paid work to the top's level and lowering the newly legible groups' precautionary buffers by about three percent, a prediction a payroll change with saving data can test.

Related literature

The most similar paper is Wang (2023), who first used the SCE's density forecasts to show that perceived wage risk lies well below the risk conventionally calibrated from panel data, even within demographic groups, and that feeding this lower, more heterogeneous risk into an incomplete-market model raises wealth inequality and the hand-to-mouth share. I share his measurement strategy, the SCE density forecast for perceived risk set against the SIPP same-job stayer panel for the realised process, and I corroborate his central finding: perceived risk lies well below the full-information benchmark in every group, and Section 4 reads the gap as an upper bound. I depart from him in four respects. First, I organise the deviation by income on the matched same-job design and show that, once decomposed, it is two-signed: the bottom over-perceives its transitory risk while the middle and the top under-perceive their permanent risk, so the cross-section sits on both sides of its own benchmark rather than uniformly below it, the reported totals still lying below throughout. Second, I document the within-person dynamics of perceived risk, its persistence and its response to realised shocks, where his evidence is cross-sectional. Third, I estimate an explicit updating rule, where he remains agnostic about belief formation by design. Fourth, I attempt to explain the gradient through a microfounded mechanism in how pay is delivered: I measure its primitive, the labelled share of transitory pay, from the itemised pay components of the same records, where he attributes the level gap to unobserved heterogeneity or overconfidence. The incidence differs accordingly: his beliefs move aggregate wealth and its dispersion, while the wedge here concentrates at the top, because the saving-relevant distortion, the under-perceived permanent risk, is itself graded by income and largest there.

The paper connects to several further strands. On subjective beliefs about income, Rozsypal and Schlafmann (2023) document an income-graded bias in expected income levels, where I study the second moment; Ben-David et al. (2018) show in the same survey that own-income uncertainty falls with income and education, and Koşar and van der Klaauw (2025) characterise the dynamics of the SCE's earnings expectations, to which this paper adds the same-job conditioning, the matched realised process, and the estimated rule. In linked Danish survey and register data, Caplin et al. (2026) likewise find survey-reported earnings risk several times below its indirectly estimated counterpart, attribute most of subjective earnings risk to job transitions, and find the risk conditional on staying in the same job small and symmetric; their instrument elicits expectations conditional on each possible transition where the question here conditions on none, so they measure the risk that comes with changing jobs and this paper the risk that remains within the job. Koşar and Melcangi (2025) estimate how the marginal propensity to consume responds to subjective earnings uncertainty, and so measure directly the margin on which the belief wedge of Section 10 appears. My proposed mechanism also speaks to the "insurance versus information" question in the partial-insurance literature (Blundell et al., 2008): Pistaferri (2001) and Kaufmann and Pistaferri (2009) use subjective expectations to separate anticipated from unanticipated shocks and show that superior information can rationalise part of the perceived-realised gap, which is the labelling channel here, made observable and graded by income.

The updating rule builds on Pischke (1995), who casts inference about permanent income as a signal-extraction problem. The under-reaction my updating rule generates is the household counterpart of the information rigidity of Coibion and Gorodnichenko (2015), which Carroll et al. (2020) carry into consumption dynamics as sticky expectations, and it runs opposite to the diagnostic over-reaction of Bordalo et al. (2018) that Sanchez and Song (2025) embed in an incomplete-markets model: the two accounts place perceived risk on opposite sides of the full-information benchmark, and the shortfall of Section 4 favours under-reaction. Experience-based learning (Malmendier and Nagel, 2011) instead grades the gain by age, where mine rises with income at fixed demographics.

Finally, the mechanism rests on documented features of how pay is delivered across the distribution, performance pay concentrated at the top (Lemieux et al., 2009), high-income shocks predominantly transitory (Guvenen et al., 2021), and volatile, fragmented pay at the bottom (Hardy and

Ziliak, 2014; Morduch and Schneider, 2017).

2 Data and measurement

The paper needs two objects that can be set against each other: what a worker believes about her wage risk, and the risk her wage actually carries. I measure the first from the New York Fed's Survey of Consumer Expectations (SCE) and the second from the Survey of Income and Program Participation (SIPP), both monthly within-respondent panels, the SIPP running over reference years 2013–2023 and the SCE over 2013–2025. I hold the two to the same conditioning, a worker who stays in the same job with the same employer for at least six consecutive months, so that any gap between them is a gap in beliefs and not in what is being measured, and I apply survey weights throughout. Respondents are sorted into three groups by median annual household income, the bottom below \$50k, the middle between \$50k and \$100k, and the top at or above \$100k.[1]

The perceived object comes from the SCE density question, which asks each respondent for the distribution of her own earnings growth over the next twelve months *"on the same job, same hours, same employer"*, reported as a ten-bin histogram in the density-forecast tradition of Engelberg et al. (2009). This conditioning is what makes the answer usable: by fixing the job, the hours, and the employer, it strips out the predictable path of pay, the part a worker varies herself, and the voluntary labour-supply variation that confounds risk estimates built on total earnings, so the elicited object is the conditional variance of the stochastic component of the wage change rather than of income. It also excludes the risk of losing or changing the job, which was the larger part of income risk found by (Caplin et al., 2026), which the model reintroduces separately through its employment block (Section 9). From each reported histogram I take the bin-midpoint variance as the perceived variance $\bar{\sigma}^2_{\text{perc}}$, exclude respondents who place all their mass in a single bin, trim at 1%/99%, and drop each respondent's reports from the first wave in which she records a job change.

The realised object comes from the SIPP, used in preference to the biennial Panel Study of Income Dynamics because it carries a unique employer identifier and records earnings every month, so a worker can be held to the same job and employer at the frequency the SCE question asks about. Per-month earnings are regular wage and salary together with bonuses, commissions, tips, and overtime, the amounts actually paid, and the self-employed are excluded. Recorded earnings also move with the hours actually worked, which the SCE question conditions away; Appendix C.7 measures what this difference in conditioning contributes to the comparison. I drop the observations in January, where the reported cross-wave change bunches spuriously, and trim squared changes at 1%/99%.

I residualise both objects on the standard demographic controls before any moment is computed like in Wang (2023): on age and its square, sex, education, state, and year-month fixed effects.

[1] The cut-offs are nominal and fixed across the sample. The SCE skews towards higher incomes, so group sizes are uneven (estimation sample: 1,531 / 7,501 / 11,331 observations). The \$50k cut-off is set where finer income brackets grow too thin to support the estimation (Appendix E.1), though a finer split might have located more precisely where the gradient forms. Both surveys sort by household income. The model of Section 9 maps the groups onto individual earning power, and Appendix C.6 re-sorts the SIPP side by own earnings, which preserves and mildly sharpens the realised gradient. The SCE reports household income only, so the belief side keeps the household sort.

This leaves the variation a worker's position in the income distribution explains, net of what her demographics and location predict. To sum up, the SCE yields a residualised perceived variance $\bar{\sigma}^2_{\text{perc}}$ and the SIPP the residualised log wage. Lastly, year-month effects absorb aggregate shocks common to all groups. The SCE alone carries a survey-tenure term that absorbs the panel conditioning of repeated subjective reports (Kim and Binder, 2023), absent from realised pay.

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3 Perceived risk by income group

Following the measurement specification that Section 2 sets out, I first look at simple facts within the data. I find that perceived risk is steeply graded by income, and the grading extends to its persistence within person, which both are facts not previously documented in literature.

Levels

Ben-David et al. (2018) document, in the same survey, that households' uncertainty about their own income growth falls with income and education. The matched-stayer conditioning and the realised counterpart here sharpen that pattern into a statement about wage risk, and so measured, perceived variance is steeply graded by income, with the bottom of the distribution far above a middle and top that sit together. As shown in Figure 1, the weighted group means are 20.6 at the bottom, 11.3 in the middle, and 11.4 at the top, so a worker at the bottom perceives about 1.8 times the wage uncertainty of one at the top. The gradient takes the form of a step, since the middle and the top are statistically indistinguishable under the midpoint measure and track each other closely, which I verified to be robust to movements in the cutoff. The bottom series does drift upwards over the sample, which may reflect the fixed nominal cut-offs or the greater cyclical exposure of low-income pay.

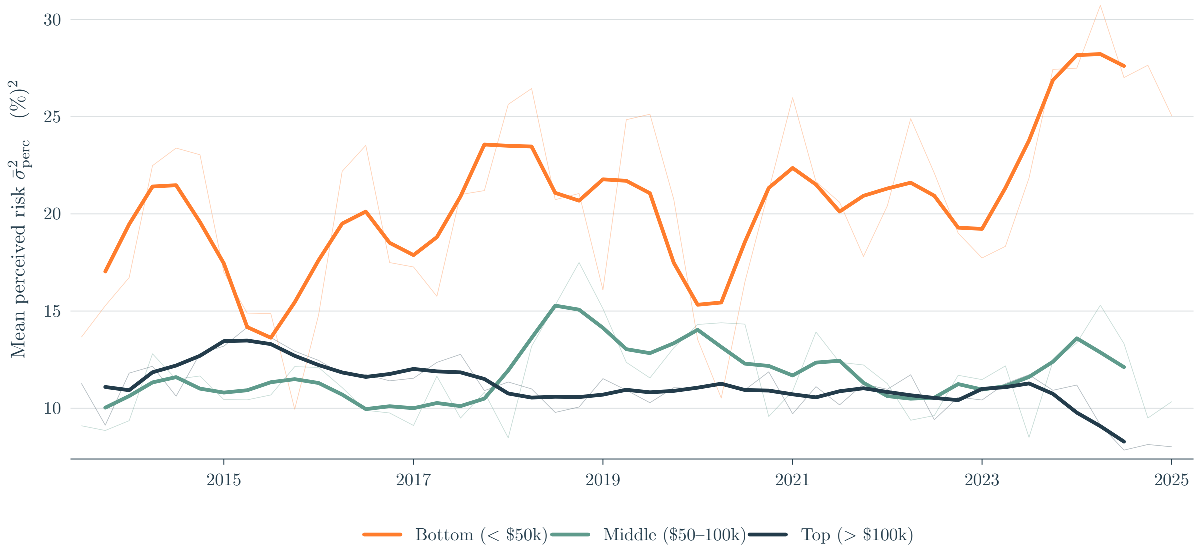

Figure 1: Group-mean perceived variance $\bar{\sigma}^2_{\text{perc}}$ by income group: quarterly means (thin) and their 4-quarter centred moving average (thick). Matched-stayer, trimmed and residualised SCE panel as in Section 2.

The bottom group is generally also the least numerate, so a natural concern is that its higher reported risk is an artefact of how it reports. The gradient, however, survives the numeracy and inflation-uncertainty controls of Appendix C.4, and the bottom uses more of the ten histogram bins than the top so its wider densities are not due to a coarser use of the histogram. If anything these controls understate the gradient, since the numeracy the bottom lacks is partly the ability to read its own fragmented pay, which the mechanism of Section 7 takes up, which is also why I keep occupation out of the baseline residualisation.

Persistence

Perceived risk is also more persistent within person at the bottom than at the top. I estimate a within-respondent AR(1) in the perceived variance $\bar{\sigma}^2_{\text{perc}}$, with persistence coefficient $\hat{\phi}_g$, by Blundell-Bond system GMM (Blundell and Bond, 1998). The short, wide SCE panel makes the within estimator severely Nickell-biased, so I instrument the lagged level with deeper lags to remove that bias, following Ma et al. (2020), who opts for this estimator for the same short-panel reason.

	Monthly		Quarterly	
Group	$\hat{\phi}_g$	SE	$\hat{\phi}_g$	SE
Bot	$+0.343^{***}$	0.046	$+0.632^{***}$	0.062
Mid	$+0.142^{***}$	0.026	$+0.594^{***}$	0.088
Top	$+0.144^{***}$	0.025	$+0.377^{***}$	0.076
Wald $(B = T)$	$p = 0.0001$		$p = 0.009$	

Table 1: Within-respondent AR(1) persistence $\hat{\phi}_g$ of perceived variance, Blundell-Bond system GMM. Instruments: $\bar{\sigma}^2_{\text{perc}}$ at lags 3–4 in differences plus the BB level moment, collapsed. Two-way clustered SEs (respondent, quarter). The same instrument set is used at both cadences.

The bottom is the most persistent of the three groups, and its equality with the top is rejected at both cadences. The gradient is not inherited from the realised process: for the persistence to come from the wage process itself, the squared wage changes would have to be more persistent at the bottom as well, and if anything they are less so, as Appendix C.5 shows. Nor is it an artefact of the cadence. Classical noise in the monthly reports biases the monthly persistence downward and partly averages out in quarters, which is why the quarterly estimates are higher for every group, but noise cannot carry the entire gap, since that would require an implausibly persistent true process, so I read the true persistence as lying between the two sets of estimates; the ranking is the same at both, with only the middle's position between the bottom and the top left unpinned. Taken with the level, the persistence leaves the assumption that perceived risk equals realised risk with two gradients to reproduce rather than one: a calibration from realised wage data would have to match both.

4 The realised wage process

If the belief gradient reflected a gradient in actual risk, it would show up in the realised wage process, and it does not. Realised shocks nonetheless move beliefs, and together these two facts pin down what a belief rule must deliver.

Wage model

I use the standard permanent-transitory decomposition (Meghir and Pistaferri, 2004):

$$w_{i,t} = z_{i,t} + e_{i,t}, \qquad e_{i,t} = p_{i,t} + \theta_{i,t}, \qquad p_{i,t} = p_{i,t-1} + \psi_{i,t}.$$

Here $w_{i,t}$ is the residualised log wage, $z_{i,t}$ the predictable Mincerian path, and $e_{i,t}$ the stochastic component that is the focus of the rest of the paper. The permanent shock $\psi_{i,t}$ is iid with variance $\sigma^2_{\psi,g}$, where g indexes the worker's income group, and accumulates through the random walk $p_{i,t}$ (a merit raise, cost-of-living adjustment); the transitory shock $\theta_{i,t}$ is iid with variance $\sigma^2_{\theta,g}$ (a discretionary bonus, unpredictable working hours).

Matching SCE to the model

The SCE question conditions on *same job, hours, employer*, which fixes $\Delta z = 0$, so the elicited variance is that of the stochastic component alone:

$$\bar{\sigma}^2_{\text{perc},i,t} \;\equiv\; \text{Var}_{i,t}(\Delta w_{i,t+1} \mid \Delta z = 0) \;=\; \text{Var}_{i,t}(\Delta e_{i,t+1}).$$

On the SIPP I residualise on demographics (removing z) and difference to recover Δe directly. The recorded wage is monthly earnings per average weekly hour reported for the month, so tips, overtime, and premium pay all enter it: the recorded counterpart of the worker's paycheck rather than of her contracted rate. Both sides measure the stochastic component of the same-employer wage change, but the hours conditioning differs, since the question holds hours fixed while the recorded wage moves with them; Appendix C.7 restricts the SIPP changes to months in which reported hours do not change and repeats the identification below on the restricted moments.

Identification of the wage primitives

The one-period change is $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$. I write $\bar{S}_{m,g}$ for the variance of this monthly change, $\mathbb{E}[(\Delta e)^2]$. Because ψ and θ are independent and iid, it has only two non-trivial moments per group:

$$\bar{S}_{m,g} = \sigma^2_{\psi,g} + 2\sigma^2_{\theta,g}, \qquad \mathrm{Cov}(\Delta e_t, \Delta e_{t-1}) = -\sigma^2_{\theta,g}.$$

The first-order autocovariance identifies $\sigma^2_{\theta,g}$ and the remaining variance identifies $\sigma^2_{\psi,g}$, so the two wage primitives are exactly identified per group; Appendix C.6 checks the zero restrictions the model imposes at higher lags and the sensitivity to the stayer conditioning.

Group	$\sigma^2_{\theta,g}$ (transitory)	$\sigma^2_{\psi,g}$ (permanent)	$\bar{S}_{m,g}$
Bot	2.52	11.89	16.92
Mid	4.42	11.60	20.44
Top	5.83	11.22	22.88

Table 2: SIPP permanent-transitory identification per income group, monthly cadence, in $(\%)^2$. Moments are computed from unrounded inputs and rounded for display.

Permanent variance is roughly flat across groups, while transitory variance rises mildly with income, plausibly the bonus pay that arrives at the top. The realised gradient therefore runs, if anything, *opposite* to the perceived one. Restricting the changes to months of unchanged reported hours sharpens rather than overturns the pattern, with both components rising mildly with income (Appendix C.7), so the contrast with the perceived gradient does not rest on the hours conditioning.

The FIRE benchmark

To compare the monthly process with the 12-month SCE horizon, I roll it forward twelve months under iid shocks (derivation in Appendix A):

$$\mathrm{Var}(e_{t+12} - e_t) \;=\; 12\sigma^2_{\psi,g} + 2\sigma^2_{\theta,g}, \tag{1}$$

the twelve-month risk a full-information rational-expectations (FIRE) agent would report, near 148 $(\%)^2$ for every group, and 79–98 under the same-hours conditioning of Appendix C.7. Perceived variance lies far below either benchmark, at 20.6, 11.3, and 11.4, between a quarter and a thirteenth of them (Table 5, Appendix A), which is the same gap documented in Wang (2023).

How much of this gap is genuine miscalibration is hard to say, because the benchmark itself is fragile. Equation (1) multiplies the permanent variance by twelve, and SIPP earnings are self-reported: the classical part of reporting error is read as transitory, but any persistent part is indistinguishable from true permanent shocks and enters the benchmark twelvefold (Bound and Krueger, 1991). Administrative payroll data, which carry no reporting error, place job-stayers' base-wage dispersion well below the benchmark (Grigsby et al., 2021). Measuring the twelve-month change directly rather than extrapolating it does not settle the level either, since the answer moves by an order of magnitude with the treatment of outliers, as Appendix C.8 shows.[2] The error is plausibly common across the groups, however, so it cancels from the cross-group comparison. I therefore read the level gap only as an upper bound and rest the argument on the contrast.

Belief dynamics and realised shocks

The cross-section rules out the realised process as the *source* of the gradient, but the realised process can still drive belief *dynamics*: Figure 2 shows that pooled perceived risk co-moves with the absolute deviation of aggregate wage growth from its sample mean, consistent with surprises to wage growth moving perceived risk over time.[3]

Figure 2: SCE perceived risk and the absolute deviation of SIPP aggregate wage growth from its sample mean, both standardised, pooled across income groups, 2015–2022.

Together these facts impose two requirements on any belief rule. It must let perceived variance move, since the dynamics rule out constant beliefs; and it must let realised shocks drive that movement. Whether the response to a largely common signal differs by group is then an estimation question, taken up within person in Section 6.

5 A belief-updating rule

In the spirit of Pischke (1995), I assume the agent cannot tell a permanent shock from a transitory one. Each period she sees a single squared pay change, which mixes the two, since $\mathbb{E}[(\Delta e)^2] = \sigma_\psi^2 + 2\sigma_\theta^2$, and she must split that surprise, reading a share λ as news about her permanent earning power and the rest as transitory. The assumption concerns the residual a worker cannot label as one or the other, and it concerns the second moment: Druedahl and Jørgensen (2020) infer from consumption responses that households track the persistent component of their income well, while what is formed here from the same monthly pay changes is a belief about its variance. The mechanism of Section 7 microfounds the assumption by granting the best-informed workers a

[2] The same-hours correction, unlike the reporting error, does not cancel across groups, the restricted benchmark rising with income where the baseline is flat; Appendix C.7 shows the cross-group comparisons survive it.

[3] Absolute deviations, since any realised shock should raise perceived uncertainty; squared deviations give a very similar plot. The comovement motivates the shock regressor of Section 6; the within-person estimates there are the evidence that realised shocks move beliefs.

cleaner separation of the labelled part of pay.

Her perceived wage risk then has two components, one permanent and one transitory, while the SCE gives one perceived-variance series per group, and one series cannot separate two moving beliefs. I therefore let the permanent belief evolve, $\hat{\sigma}^2_{\psi,t}$, and hold the transitory at a static prior, $\sigma^2_{\theta,\text{perc}}$, since the dynamics belong with the belief saving responds to and a static prior is the natural home of an unresolved reading error, recorded exactly as a transitory shock would be. The data do not identify the assignment: the series is equally consistent with a static permanent bias and a moving transitory belief, and any misallocation between the two loads onto λ.

The permanent-variance belief follows an exponentially weighted moving average (EWMA) of past squared shocks, with the attribution share as a separate weight:

$$\hat{\sigma}^2_{\psi,t} \;=\; \underbrace{(1-\mu)}_{\text{persistence}}\;\hat{\sigma}^2_{\psi,t-1} \;+\; \underbrace{\mu}_{\text{update weight}}\;\underbrace{\lambda}_{\text{attribution share}}\;(\Delta e_{t-1})^2. \tag{2}$$

Here μ is the update speed, so a higher μ moves the belief more on each period's news, and λ is the share of each squared surprise the agent reads as informative about her permanent earning power. Since $(1-\mu)$ and μ sum to one, the recursion is analogous to an integrated GARCH(1,1) in the squared surprise; it is also the steady-state Kalman filter for an unknown permanent variance modelled as a random walk, with μ the Kalman gain (Appendix D.1 gives the moment-matched filter and sets the gain by the signal-to-noise ratio), and applied period after period the constant gain is the perpetual-learning rule for a drifting variance. Section 8 takes up what departure from full information her constant attribution represents, once Section 7 grades it by legibility.

Substituting the belief into the FIRE identity (1) gives the level the agent reports:

$$\bar{\sigma}^2_{\text{perc},t} \;=\; 12\,\hat{\sigma}^2_{\psi,t} \;+\; 2\sigma^2_{\theta,\text{perc}}. \tag{3}$$

Strict FIRE is the corner $\mu = 0$ with both beliefs at the truth; it implies a constant reported variance, which the dynamics of Section 3 rule out. Within the learning family ($\mu > 0$) FIRE survives only as a long-run average: taking unconditional expectations of (2) in a stationary state gives $\mathbb{E}[\hat{\sigma}^2_{\psi}] = \lambda\,\bar{S}_m$, so the belief is right on average exactly when $\lambda^* = \sigma^2_{\psi}/\bar{S}_m$, which Table 2 puts at 0.70, 0.57, and 0.49 across the three groups; the transitory counterpart is $\sigma^2_{\theta,\text{perc}} = \sigma^2_{\theta}$. The estimates of Section 6 test the prior condition directly, and the attribution condition only through its cross-group ratios.

The identity's role in the estimation is to supply the dynamics: substituting it into the recursion delivers the observable equation of Section 6, whose persistence, news response, and intercept identify the rule. It is not inverted to recover the level of the permanent belief from a reported level, because Section 4 reads the level gap between the surveys as an upper bound: the shortfall sits in the term the identity compounds twelve times, the part of the answer a respondent must accumulate over a year and plausibly does not, while the intercept carries no compounding and crosses the surveys on the scale of the realised process (Appendix D.3). The permanent belief the model carries therefore comes from the mechanism, in the units of the realised process (Section 8).

6 Estimation

The belief recursion (2) is written in the unobserved $\hat{\sigma}^2_\psi$. Combining it with the level identity (3) eliminates the belief and leaves an AR(1) in the observable perceived variance:[4]

$$\bar{\sigma}^2_{\mathrm{perc},i,t} = \underbrace{2\mu_g\,\sigma^2_{\theta,\mathrm{perc},g}}_{\beta_{\mathrm{cons}}} + \underbrace{(1-\mu_g)}_{\beta_{\mathrm{lag}}}\,\bar{\sigma}^2_{\mathrm{perc},i,t-1} + \underbrace{12\mu_g\lambda_g}_{\beta_{\mathrm{shock}}}\,(\Delta e_{i,t-1})^2 + \alpha_i + u_{i,t}, \tag{4}$$

with α_i a person fixed effect. The three coefficients map one-to-one to the structural parameters, with the person effects normalised to mean zero within each group: $\mu_g = 1 - \hat{\beta}_{\mathrm{lag}}$, $\sigma^2_{\theta,\mathrm{perc},g} = \hat{\beta}_{\mathrm{cons}}/(2\mu_g)$, and $\lambda_g = \hat{\beta}_{\mathrm{shock}}/(12\mu_g)$. Each is identified from a distinct feature of the data: β_{lag} from the within-respondent persistence, β_{cons} from the group level once the news loading is netted out, and β_{shock} from the response to the lagged realised shock.

The shock regressor

The rule calls for the agent's own $(\Delta e_{i,t-1})^2$. The SCE contains no individual wage histories, so I substitute the SIPP cell mean over workers in the same cell c = (group × Census division × month), following Guerreiro (2023); workers in the same local labour market share components of wage news, transmitted through co-workers and local media and salient for expectations (Kuchler and Zafar, 2019). The cell series is smoothed with a 3-month trailing MA.

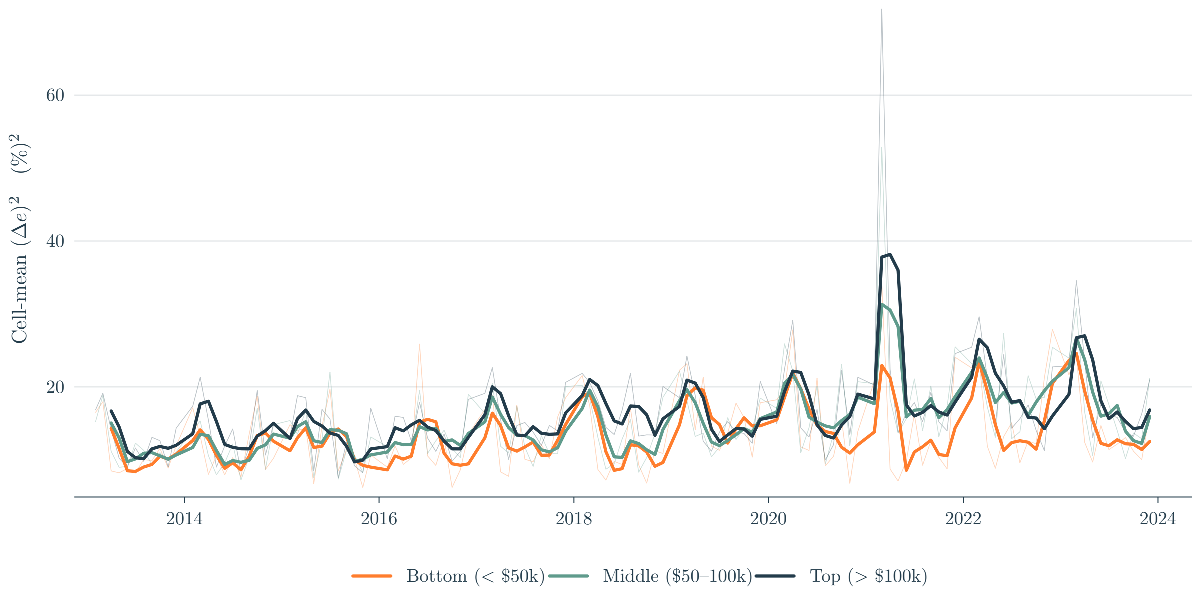

Figure 3: Cell-mean $(\Delta e_{i,t})^2$ by income group, national monthly, in $(\%)^2$. Thin lines: raw monthly cell means. Thick lines: the 3-month trailing MA used as the regressor in (4).

[4]Solving (3) for the belief gives $\hat{\sigma}^2_{\psi,t} = (\bar{\sigma}^2_{\mathrm{perc},t} - 2\sigma^2_{\theta,\mathrm{perc}})/12$; substituting both belief terms into (2), multiplying by 12, collecting the intercept $(2\mu\sigma^2_{\theta,\mathrm{perc}})$, and adding group subscripts, a person fixed effect α_i, and some iid error $u_{i,t}$ gives (4). The intercept and the person effects are identified only jointly, since a constant can move between them, so the normalisation $\mathbb{E}[\alpha_i \mid g] = 0$ fixes the split: the group-average level of reported variance enters β_{cons} and identifies the prior, while person-specific deviations, including any stable reporting style, enter α_i. Whatever mean reporting width a group carries therefore loads into its prior, the channel Appendix C.4 bounds.

Figure 3 shows that the shocks are largest for the two upper groups, consistent with their higher $\sigma^2_{\theta,g}$ and with the concentration of discretionary pay at the top (Guvenen et al., 2021; Halvorsen et al., 2024). The bottom's shocks are, if anything, the smallest, so any stronger belief response there reflects how the bottom updates rather than the size of the shocks it receives.

Replacing the own shock with a cell mean is a measurement-error problem that attenuates β_{shock}, and hence $\hat{\lambda}$, towards zero: the cell mean tracks only the common component of a worker's shock, so the slope recovers her response up to a group-level loading, which Appendix D.3 measures at 0.43, 0.56, and 0.58 from the bottom up, and correcting by these widens rather than narrows the cross-group gradient. What the coefficient measures directly is therefore the response of a belief to common, cell-level wage news, and I read only its cross-group pattern, never its level. Little more is asked of it: the model of Section 9 carries the attribution implied by the measured labelled share, and the estimated coefficient enters only the ratio comparison of Appendix D.3.

Estimates by income group

Joint GMM on (4) by income group yields the estimates in Figure 4, set against the realised SIPP transitory variance. The shock is treated as endogenous, since it shares a survey-timing and aggregate-news channel with the belief, and I instrument with its own lags and lags of the perceived variance. The full estimates, with standard errors, the Hansen J, and sample sizes, are in Table 6 (Appendix C).

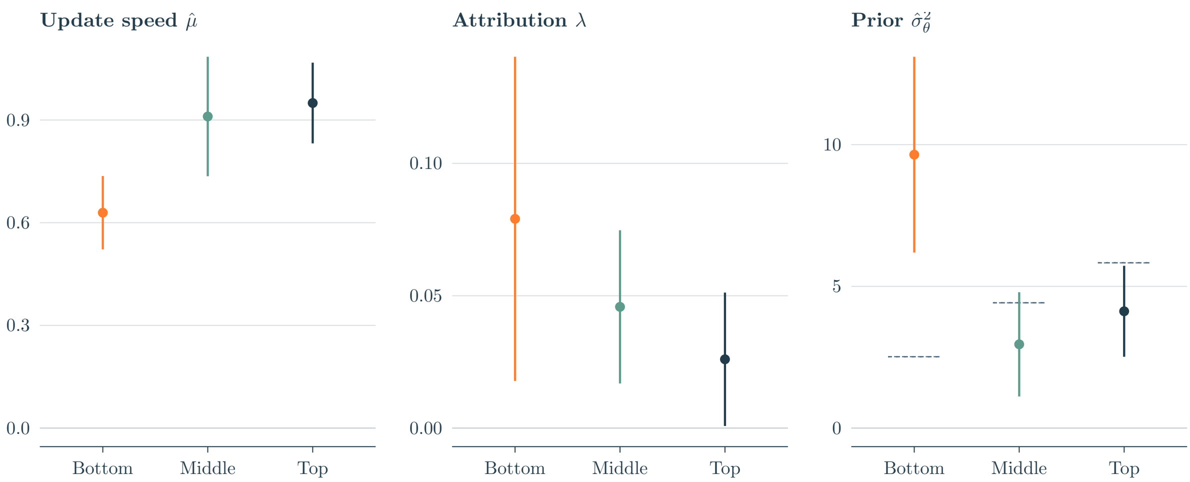

Figure 4: Headline estimates by income group: update speed μ_g, attribution λ_g, and transitory prior $\sigma^2_{\theta,\text{perc},g}$, with 95% confidence intervals. Grey dashes in the prior panel mark realised transitory variance $\sigma^2_{\theta,g}$ (SIPP).

Update speed. The bottom updates significantly more slowly than the middle or top, matching the persistence ordering of Table 1: the influence of a given wage shock on its belief decays more slowly.

Attribution. Each $\hat{\lambda}_g$ is positive and significant, and the point estimates fall monotonically with income, the bottom's at three times the top's. The decline is not statistically sharp, as cross-group equality is not rejected, and part of it is rational, since the full-information share λ^*_g itself falls with income, at about half the estimated pace (Section 5).

Transitory prior. Realised transitory variance rises with income; the perceived prior does the opposite. The bottom's prior sits almost four times its realised value, while the middle and top lie near or below theirs. The bottom is the only group whose prior is materially detached from its realised data, and the gradient is significant. The inflated prior is therefore the dominant driver of the *level* of the bottom's perceived variance, separate from the news channel.

Robustness

Adding an occupation × year-month fixed effect to the shock construction preserves the persistence and attribution gradients and sharpens the latter, while the prior gradient holds in direction but loses cross-group significance, as Appendix C.1 reports. I do not adopt it as the baseline, since low-income workers sort into structurally more uncertain occupations (Morduch and Schneider, 2017), so partialling out occupation removes part of the channel.

The estimates change little when the instrument lags, the trim, and the COVID handling vary one at a time (Appendix C.2): the persistence gradient survives every variant except the tighter within-group trim, where its significance turns marginal, and the attribution gradient stays monotone throughout. The bottom's prior excess holds in every variant, 6.4 to 11.2 against a realised 2.52, though its cross-group significance fails under the COVID exclusion and the occupation control. The COVID years are realised wage variation that the SIPP benchmark sample shares, so excluding them discards identifying variation; the occupation control over-controls for the reason above. The upward drift of the bottom series, noted in Section 3, matters as little: a linear detrend or calendar-year effects leave the bottom's prior and gain within decimals of their canonical values and the cross-group Walds intact (Appendix C.2).

Lastly, splitting the shock by sign, with the groups pooled for power, puts about twice the weight on upside news as on downside, an asymmetry I read as suggestive only (see Appendix C.3 for a more in-depth discussion).

7 A pay-delivery mechanism

The estimates of Section 6 give three parameters for each income group, the update speed, the attribution, and the transitory prior, each fitted freely, so the gradients across groups remain unexplained. I trace them to one feature of how pay is delivered: how legibly a worker's pay separates into a permanent and a transitory part. Legibility governs two forces of opposite sign, and because the SIPP itemises the components of a paycheck, it can be measured directly, on the same stayer panel that supplies the realised process.

The legibility of pay, and the bottom's prior

The mechanism has to explain the transitory prior: the bottom's is 9.65 against a realised transitory variance of 2.52, while the middle's and the top's are 2.96 and 4.12 against realised values of 4.42 and 5.83, so the prior lies above realised risk at the bottom and below it higher up. Explaining both sides requires two forces, one that lowers the prior and one that raises it, and I conjecture that how pay is delivered decides which dominates.

The downward force: labelling. At the top, transitory pay arrives labelled: a bonus or a commission comes on its own line, a furlough comes with a return date, and there is a single salary to read it all against, so the worker nets the transitory part out before she updates. At the bottom the same pay arrives scattered, as variable shifts, cash tips, and supplements layered on uncertain hours, and a downturn arrives ambiguous, a sector closing or gig work drying up rather than a furlough with an end date, so she reads the whole change as one surprise. Delivery grades this legibility by income: performance pay concentrates at the top and arrives on separate line items (Lemieux et al., 2009), high-income downturns arrive in classifiable forms (Hoynes et al., 2012), and low-income pay is fragmented where high-income pay is a single salary (Hardy and Ziliak, 2014; Morduch and Schneider, 2017). Let $\rho_g \in [0, 1]$ be the share of transitory pay a worker can separate and net out; only the unlabelled residual feeds the recursion (2), so her prior falls to $(1 - \rho_g)\sigma^2_{\theta,g}$, the netting floor. The floor is reached only if labelling resolves pay before it arrives, an itemised but still uncertain commission carrying its variance all the same, so I read labelled pay as largely anticipated, a bonus announced ahead of its payment, a commission earned on a known schedule; whether workers in fact net it out is tested below, where the estimated priors are set against their netting floors.

The upward force: reading noise. A worker who has netted out what arrives labelled still has to read the remainder, and at the bottom the remainder is the fragmented part of pay itself, so she cannot tell cleanly how large a change is or how much of it will last; consistent with this, within-person hours volatility falls monotonically with income, as Appendix D.5 documents. The question's conditioning falls unevenly on this problem: each respondent is asked for her risk holding hours fixed, and complying requires the separation that a salary against a fixed schedule makes easy and fragmented pay makes hard, so the worker least able to separate is the one the instruction asks most of. Hours enter on two margins that should be kept apart: a change in the reported schedule is rare and persistent, earning-power risk that the difference moments assign to $\sigma^2_{\psi,g}$ and that Appendix C.7 nets out of the realised process, while the movement of hours worked around an unchanged schedule is high-frequency and transitory, and it is through this movement that she must read her remaining pay. Whether the residual the bottom reports is unresolved reading or a truthful report of hours risk the question asks her to set aside, the moments cannot say (Appendix C.7 bounds the two); either way it enters the transitory prior and not the permanent belief. Her reading of the netted pay therefore carries an error,

$$\tilde{y}_t \;=\; p_t + \theta^u_t + \nu_t, \qquad \nu_t \overset{\text{iid}}{\sim} (0,\, \omega^2_g), \tag{5}$$

with p_t her permanent component, θ^u_t the unlabelled transitory component with variance $(1 - \rho_g)\sigma^2_{\theta,g}$, and ν_t the reading error; the structure is again that of Pischke (1995), extraction of the permanent component from an observation contaminated by transitory variation, with the reading error as a second contaminant.[5] An unresolved reading is recorded as risk, and the prior settles

[5]Recorded SIPP earnings capture the amounts actually paid, so ν never enters the moments of Section 4. The differenced error $\Delta\nu_t$ has variance $2\omega^2_g$, first autocovariance $-\omega^2_g$, and none beyond, exactly the signature of a differenced transitory shock, which is why an unresolved reading is recorded as transitory risk and why the prior is its natural home (Section 5).

above its netting floor by exactly the variance of the error,

$$\sigma^2_{\theta,\mathrm{perc},g} \;=\; (1-\rho_g)\,\sigma^2_{\theta,g} \;+\; \omega^2_g, \qquad \omega^2_g \geq 0. \tag{6}$$

Measuring the labelled share

The SIPP records, for each worker and month, her total earnings and the dollar amount of each pay component, bonuses, commissions, tips, and overtime, on the same-job stayer panel of Section 2. Within each spell I demean total pay and scale by spell-mean earnings, so what remains is the month-to-month movement of pay as a fraction of typical pay; the labelled part of that movement is the pay that arrives on its own line, bonuses and commissions, with tips and overtime counting as unlabelled, and ρ_g is the person-weighted regression share of the labelled movement in the total (Appendix D.2 gives the formula and the specification battery). The regression puts the labelled share at 0.121 at the bottom, 0.178 in the middle, and 0.358 at the top, and the gradient holds across the six robustness specifications. The shares are conservative, because the demeaning counts the slow drift of earning power as movement: a mid-spell raise puts every earlier month below the spell average and every later month above it, inflating the denominator of the share while the labelled numerator is untouched, so the labelled share of the transitory movement alone is higher, and removing a within-spell trend, which strips the drift, raises the shares to 0.176/0.256/0.488, the upper edge of the measurement band.

The share measures what arrives labelled, not what a worker reads: the split of components comes from the pay records alone, fixed before any belief moment enters, and whether she uses the labels is tested rather than assumed, since a labelled share that went unread would leave the estimated priors above the floors it implies. At the measured shares the netting floors $(1-\rho_g)\sigma^2_{\theta,g}$ are 2.21/3.63/3.74, and the middle and the top priors sit on them to within a standard error; at the bottom the prior sits 7.43 $(\%)^2$ above its floor, and the excess identifies the reading noise ω^2 of (6) (Appendix D.3). The step from a labelled dollar to a netted one still carries assumptions the pay records cannot test, about what a worker anticipates and about the independence of the labelled and unlabelled parts, so this is a structural reading of the delivery gradient, and its test is the out-of-sample variation in pay format that the conclusion describes.

The update speed and the attribution

With the labelled share measured, the update speed and the attribution can be deduced analytically and compared against the estimates. Both move with legibility through the same channel: labelling shrinks the part of a pay change the worker must still read, so a more legible worker faces a cleaner but smaller signal.

The update speed. How fast a worker updates depends on how clean a signal each pay change is: each squared change is one noisy reading of her permanent variance, the permanent shocks the object she tracks and the unlabelled transitory part the noise around them. Labelling removes part of that noise, so a more legible worker gets sharper readings and should update faster (the filter and the algebra are in Appendix D.1). The filter predicts correctly that the bottom updates slowest, but it underpredicts the spread, with implied gains of 0.76/0.78/0.82 against the estimated 0.63/0.91/0.95 (Appendix D.4). I therefore use the estimated gains in the model, and keep the filter only as a check on their ordering.

The attribution share. The attribution falls with legibility for a related reason. The rule has the agent book a single share λ of the whole squared pay change as permanent, but she has already set the labelled part aside as transitory and reacts only to the unlabelled remainder. The labelled part rides along in the change she sees and carries no news for her belief, so the more of her pay arrives labelled, the larger the inert share of each change and the smaller the fraction of the whole she books as permanent. Under joint Gaussianity of ψ and θ this gives a closed form, derived in Appendix B,

$$\lambda_g = \frac{\sigma^2_{\psi,g}\left(\sigma^2_{\psi,g} + 2(1-\rho_g)\sigma^2_{\theta,g}\right)}{(\sigma^2_{\psi,g} + 2\sigma^2_{\theta,g})^2}, \tag{7}$$

which falls in ρ_g and equals the FIRE benchmark $\lambda^*_g = \sigma^2_{\psi,g}/\bar{S}_{m,g}$ at $\rho_g = 0$. At the measured shares it is 0.677/0.524/0.401, meaning it declines with income as the λ panel of Figure 4 does.

The comparable moments are the cross-group ratios, since the cell-mean regressor attenuates the estimated level by the loading of Section 6, and correcting by the measured loadings widens rather than narrows the estimated gradient, so differential reliability does not manufacture the comparison (Appendix D.3). Measured delivery predicts bottom-to-top and middle-to-top ratios of 1.69 and 1.31, against estimated ratios of 3.04 and 1.76 with log standard errors of 0.63 and 0.59: each prediction sits within one standard error of its estimate, at about half of the estimated point gradient. Delivery therefore explains roughly half of how attribution declines with income, and the remainder is either an under-reaction beyond what labelling accounts for or sampling noise in ratios this wide.

The profile the model carries

The carried share. The share the model carries is the belief block's one fitted parameter. I pin it inside the measured band of Appendix D.2, choosing the group shares that reconcile the banded measurements with the estimated attribution ratios through (7); the reconciliation selects $\rho_g = 0.119/0.197/0.369$, close to the measured centres and well inside the measurement band.[6] The reconciled shares enter the model and nothing else: the floors, the attribution ratios, and the gain diagnostic are all evaluated at the measured shares, so the tests of the mechanism stand on the pay records alone and the reconciliation moves only what the model carries. The transitory prior is carried alongside at its estimate where it clears its netting floor and at the floor where it does not, which binds only for the middle.

From three groups to a profile. The model of Section 9 lets a worker's income drift continuously, so it needs the belief objects as functions of income rather than three points. I interpolate the carried share and the gain in levels and the prior in logs through the three anchors with monotone interpolants, holding each flat beyond the measured range and censoring the prior below at the local netting floor $(1-\rho(y))\,\sigma^2_{\theta,g}(y)$; the attribution and the permanent belief then follow from the closed forms at each income, as Appendix E.1 sets out. The prior's dip at the middle survives, since

[6] The fit is overidentified, five moments against three parameters, and the reconciliation diagnostic does not reject ($J = 1.57$ on two degrees of freedom, $p = 0.46$): the pay records, the belief responses, and the prior floors agree within sampling error. The diagnostic is a consistency check rather than a validation; the mechanism's tests are the scorecard of Appendix D.3 and the out-of-sample delivery variation the conclusion describes.

the floors are non-monotone in the same way. Figure 5 draws the measured share, the identified reading noise, and the prior they compose against income, and Table 11 in Appendix E.1 evaluates the mapping at representative incomes.

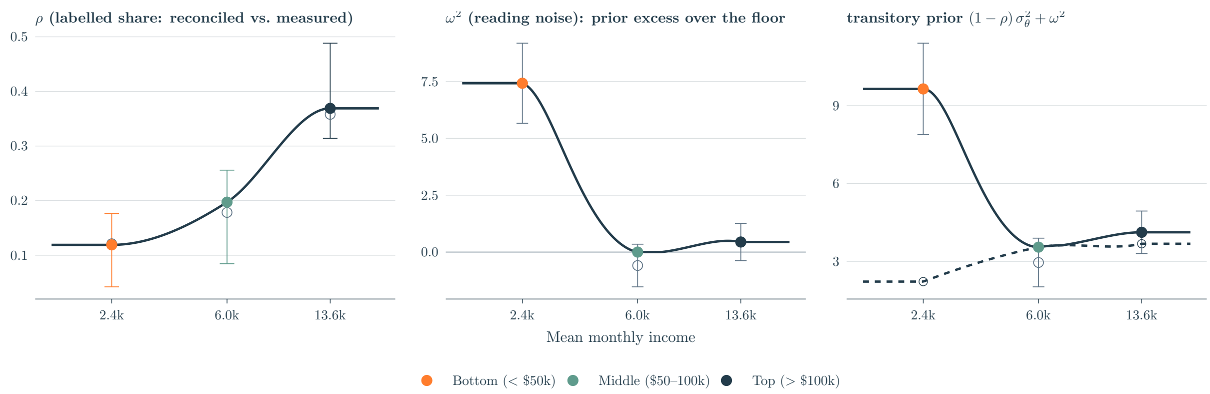

Figure 5: The labelled share ρ, the reading noise ω^2, and the transitory prior they compose (6), by income. Left: the share the model carries (filled points, with its income interpolant) against the measured share (open points), with whiskers spanning the specification band of Appendix D.2. Middle: the reading noise, the prior's excess over its netting floor, with ± 1 standard-error bands in grey; it is significant only at the bottom. Right: the carried prior (solid interpolant) as the netting floor (dashed) plus the reading noise, against the raw estimates and their ± 1 standard-error bands in grey. In $(\%)^2$ except the dimensionless share.

8 What the estimated mechanism implies analytically

Section 7 measured a single pay-delivery primitive, the legibility ρ_g of a worker's pay, and carried the transitory prior alongside it as an estimated belief moment. What the two imply for behaviour can be worked out analytically, before the calibrated model of Section 9 supplies the magnitudes.

The mechanism departs from full information in two ways. The worker nets out the labelled share of her transitory pay, about which she is unbiased given her information: she knows more than the econometrician rather than mis-perceiving the truth. She also holds her attribution fixed as netting cleans her signal, which is an under-reaction: the cleaner signal warrants a larger attribution to permanent risk than she applies, and her perceived permanent variance settles below the truth. Only the under-reaction creates a wedge in saving behaviour.

The assumed behaviour has precedents in two literatures. The constant gain itself, the same steady-state Kalman weight applied to each new surprise (Section 5), is the perpetual-learning rule for a drifting parameter, the exponentially weighted forecast that Muth (1960) showed optimal when the tracked object follows a random walk and that anchors the adaptive-learning literature (Evans and Honkapohja, 2001; Malmendier and Nagel, 2016). The binding half is the constant attribution, and it is the updating pattern the experimental literature documents: agents compress the informativeness of their signals, updating too much on weak ones and too little on strong ones (Edwards, 1968; Augenblick et al., 2025), and noisy cognition generates this compression directly (Enke and Graeber, 2023). The same pattern organises this cross-section: the bottom revises its permanent belief strongly on a signal too fragmented to warrant it, while the top under-reads the sharp signal netting leaves behind.

I proceed in three steps. I first write the perceived second moments in closed form, isolate the piece that drives precautionary saving, and collect the comparative statics in legibility. I then define the gap between perceived and true risk that organises the rest of the paper, and show that, as an error in a variance, it reaches behaviour through prudence, where a level error would act through the intertemporal-substitution motive. I finally let the belief carry its own dynamics, so that a shock to the level of income and a shock to its risk leave wedges of very different durations.

The permanent belief in closed form

Averaging the constant-gain recursion (2) over a stationary state cancels the persistence terms and leaves $\mathbb{E}[\hat{\sigma}^2_\psi] = \lambda\, \bar{S}_{m,g}$, the attribution times the variance of the surprise she updates on. To rest at the truth she would need $\lambda^*_g = \sigma^2_{\psi,g}/\bar{S}_{m,g}$, the genuine permanent share of the full monthly change at steady state. Netting changes that surprise: setting the labelled transitory pay aside leaves a residual of smaller variance $V_{u,g} = \sigma^2_{\psi,g} + 2(1-\rho_g)\sigma^2_{\theta,g} < \bar{S}_{m,g}$ whose permanent part is still $\sigma^2_{\psi,g}$, so the residual is a cleaner signal, and a Bayesian agent would raise her attribution to $\sigma^2_{\psi,g}/V_{u,g} > \lambda^*_g$ and stay unbiased. The constant-gain agent I assume in Section 5 does not: she keeps her attribution at the benchmark λ^*_g as netting cleans the signal, and her belief settles at

$$\mathbb{E}[\hat{\sigma}^2_\psi] \;=\; \lambda^*_g\, V_{u,g} \;=\; \sigma^2_{\psi,g}\,\frac{V_{u,g}}{\bar{S}_{m,g}}. \tag{8}$$

The discount $V_{u,g}/\bar{S}_{m,g} \le 1$ is the under-reaction, the weight she applies as a fraction of the weight the cleaner signal warrants. The weight is exactly one when nothing is labelled ($\rho_g = 0$, so $V_{u,g} = \bar{S}_{m,g}$) and falls as she nets more, so the more legible her pay the further her perceived permanent risk sits below the truth.

The same belief can equally be written as an attenuated attribution on the full change she observes: $\lambda = \sigma^2_{\psi,g}V_{u,g}/\bar{S}^2_{m,g} = \lambda^*_g(V_{u,g}/\bar{S}_{m,g})$ of (7) sits below the benchmark λ^*_g and rests at the same belief, since $\lambda\,\bar{S}_{m,g} = \lambda^*_g V_{u,g}$. The labelled part rides in the change she reacts to and carries no news about permanent risk, so it attenuates her attribution exactly as classical measurement error attenuates a regression coefficient.[7] The estimation of Section 6 does not establish the wedge on its own. A Bayesian agent who netted out the labelled part of her pay and raised her attribution to $\sigma^2_{\psi,g}/V_{u,g}$ would rest at the unbiased $\sigma^2_{\psi,g}$, and in the estimating equation (4) she is nearly indistinguishable from the under-reacting agent: the coefficient her rule implies on the observed squared change differs from the netting reading of Appendix B only by the factor $V_{u,g}/\bar{S}_{m,g}$, moving the predicted ratios from 1.69 and 1.31 to 1.99 and 1.48, a distinction the standard errors cannot resolve.

The two rules separate in the gain instead. The Bayesian's attributed signal has variance $2(\sigma^2_{\psi,g})^2$, nearly flat across groups, so at any common drift her steady-state gain is nearly flat as well, at about 0.8 for every group when the drift is fitted to the estimated gains; the estimated gains instead rise from 0.63 to 0.95 and reject equality of the bottom and the top decisively (Table 6), the rising order the constant-attribution filter predicts, its observation noise falling with legibility. The wedge therefore rests on the constant attribution assumed in Section 5, and the evidence is

[7] A third reading is the implemented one: the agent updates on the squared change of her own noisy netted reading (5), whose mean is $V_{u,g} + 2\omega^2_g = \sigma^2_{\psi,g} + V_x$ with $V_x = 2\sigma^2_{\theta,\mathrm{perc},g}$, and the recursion's gain is the rescaled $\tilde{\lambda} = \sigma^2_{\psi,g}V_{u,g}/[\bar{S}_{m,g}(\sigma^2_{\psi,g} + V_x)]$, chosen so that the resting mean is unchanged (Appendix D.1, Table 11): the reading error inflates her signal and the rescaling offsets it, so it never reaches the resting permanent belief.

consistent with that reading rather than decisive for it, since a Bayesian free to set her drift group by group can match any profile of gains (Appendix D.4).

The prior does not enter (8), though its excess over realised transitory risk is largest at the bottom. Averaging the recursion over a stationary state leaves the resting belief at $\tilde{\lambda}_g \, \mathbb{E}[x_t]$ whatever the gain, and the prior sits in the denominator of the rescaled attribution and in the mean of the signal, so it cancels between the two: the reading error that inflates it survives in $\sigma^2_{\theta,\text{perc},g}$ and never in the resting permanent belief. The cancellation is a property of the rescaling I adopt, under which the attribution tracks the mean of the signal the agent actually reads, and the gain is prior-neutral for the same reason (Appendix D.1). The permanent belief that drives precautionary saving therefore moves with legibility alone, and the saving distortion follows the legibility and not the prior.

The perceived-risk wedge

Define the perceived-risk wedge as the gap between the agent's perceived second moments and the true ones, summed across both shocks,

$$\Delta\sigma^2 \;=\; 12\big(\hat{\sigma}^2_{\psi} - \sigma^2_{\psi,g}\big) \;+\; 2\big(\sigma^2_{\theta,\text{perc}} - \sigma^2_{\theta,g}\big), \tag{9}$$

twelve times the permanent gap plus twice the transitory gap, the weights with which the two shocks enter the twelve-month variance of (1). Its two pieces are signed differently: the permanent gap $\hat{\sigma}^2_{\psi} - \sigma^2_{\psi,g} = \sigma^2_{\psi,g}[V_{u,g}/\bar{S}_{m,g} - 1] \le 0$ is one-signed, since legibility can only depress perceived permanent risk, while the transitory gap $\sigma^2_{\theta,\text{perc}} - \sigma^2_{\theta,g} = \omega^2_g - \rho_g \sigma^2_{\theta,g}$, by (6), is two-signed, raised by the reading noise at the bottom and lowered by netting higher up. Because the two gaps pull in opposite directions, perceived total risk can sit on either side of the truth, and across groups it crosses over: the prior excess dominates at the bottom, where the wedge is positive, and legibility dominates higher up, where it turns negative and reflects information rather than error. Figure 6 draws the wedge at the measured shares ρ_g, in the benchmark's own units rather than the reported level of Section 3: the two-sided prior of Section 7 reappears as a property of $\Delta\sigma^2$, the total crosses the benchmark only faintly, and the sharp crossing comes from the transitory part alone.

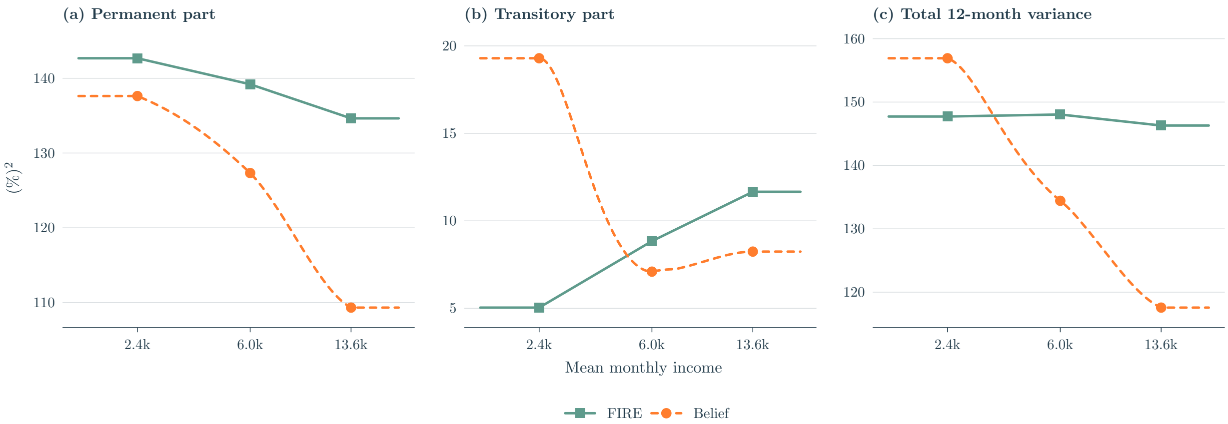

Figure 6: Perceived twelve-month risk against the FIRE benchmark, in closed form along the measured belief profile of Figure 5. Each panel draws the benchmark (solid) against its perceived counterpart (dashed), and the distance between the lines is the matching piece of the wedge (9): the permanent part $12\sigma^2_{\psi,g}$ against $12\hat{\sigma}^2_{\psi}$ in (a), the transitory part $2\sigma^2_{\theta,g}$ against $2\sigma^2_{\theta,\text{perc}}$ in (b), and the total in (c). Points mark the three income groups. In $(\%)^2$.

A gap in second moments reaches behaviour through the curvature of the value function. The continuation value evaluated under the agent's perceived income density rather than the true one differs only through their spread, since the two share a mean, and each shock enters the problem through its own argument, the permanent innovation through the growth factor and the transitory draw through next-period resources, and the two are independent. A second-order expansion then leaves one curvature term per shock,

$$\widetilde{\mathbb{E}}\,V - \mathbb{E}\,V \;\approx\; \tfrac{1}{2}\left(\hat{\sigma}^2_{\psi} - \sigma^2_{\psi,g}\right) V_{\psi\psi} \;+\; \tfrac{1}{2}\left(\sigma^2_{\theta,\text{perc}} - \sigma^2_{\theta,g}\right) V_{\theta\theta}, \tag{10}$$

the monthly variance gap of each shock times the curvature of the continuation value in that shock, both derivatives taken in the level of each innovation at its mean of one, which the mean-one normalisation holds fixed as the variance moves, where $\widetilde{\mathbb{E}}$ and $\mathbb{E}$ are the expectations under her perceived and the true density. The twelve-and-two weights of the wedge (9) belong to the reported object at the survey horizon and not to the Bellman equation, which is monthly. As a second-order approximation the expansion locates the channel and leaves its magnitude to the model of Section 9; the argument that follows rests on the prudence term of the Euler equation, to which this expression is the compact counterpart.

The comparative statics of the next subsection make precise which part of the belief block carries the wedge, and the two subsections after it trace the wedge through prudence and the belief's own dynamics.

Comparative statics in legibility

Every object in the belief block moves through the unlabelled residual variance $V_{u,g}$, which legibility lowers ($\partial V_{u,g}/\partial \rho_g = -2\sigma^2_{\theta,g}$). Differentiating the permanent belief (8), the gain through its observation noise $R_g = 2(\sigma^2_{\psi,g})^2(V_{u,g}/\bar{S}_{m,g})^2$, derived in Appendix D.1, and the attribution (7) through $V_{u,g}$ gives the signs collected in Table 3, where $g_p \equiv \hat{\sigma}^2_{\psi} - \sigma^2_{\psi,g}$ is the saving-relevant gap.

	Response to		
Belief object	ρ_g	ω_g^2	Margin it reaches
Permanent belief $\hat{\sigma}_\psi^2$	$-$	0	saving, through prudence
Saving-relevant gap $g_p \equiv \hat{\sigma}_\psi^2 - \sigma_{\psi,g}^2$	$-$	0	saving
Attribution λ_g	$-$	0	news response
Update gain μ_g	$+$	0	belief dynamics
Transitory prior $\sigma_{\theta,\text{perc},g}^2$	$-$	$+$	buffered, second order

Table 3: Comparative statics of the beliefs and the estimated rule in legibility ρ_g and the identified reading noise ω_g^2 of (6); entries are the signs of the partial derivatives. The zeros in the ω_g^2 column hold for the resting belief, where the recursion's gain absorbs ω_g^2 by construction (Appendix D.1); in the response to a risk shock the prior re-enters through the gain (Section 11).

Legibility moves every object in the table: it raises the update gain and lowers the attribution, so it governs how beliefs evolve and how they respond to news, and it is also the only force holding perceived permanent risk below the truth. The reading noise enters only the transitory prior, one for one, and leaves the signal-to-noise ratio, the attribution, and the resting permanent belief untouched, since the recursion's own gain absorbs it by construction, as Appendix D.1 shows. The saving-relevant gap therefore depends on legibility alone: two groups that read their pay equally legibly carry the same gap however differently their priors are pinned, so the behavioural wedge widens with legibility along the income gradient and not with the prior.

Prudence and the incidence of the wedge

A belief about a variance moves saving through a different part of preferences than a belief about the level would. The saving choice is disciplined by the Euler equation, which weighs marginal utility today against its expected value next period, so a belief about future risk reaches saving only through the expected marginal utility $\widetilde{\mathbb{E}}[u'(c')]$, where $\widetilde{\mathbb{E}}$ and $\widetilde{\text{Var}}$ are the expectation and variance under the agent's belief $\hat{\sigma}_\psi^2$. Expanding it around mean next-period consumption, $\widetilde{\mathbb{E}}[u'(c')] \approx u'(\widetilde{\mathbb{E}}c') + \frac{1}{2}u'''(\widetilde{\mathbb{E}}c')\,\widetilde{\text{Var}}(c')$, a belief about the mean would move the first term through u'', the curvature that fixes the elasticity of intertemporal substitution, while a belief about the variance touches only the second, through u'''. A perceived-variance error $\hat{\sigma}_\psi^2 - \sigma_{\psi,g}^2$ therefore acts on saving through prudence, just as a level error would act through the elasticity of intertemporal substitution. Carrying the consumption policy through,

$$\widetilde{\mathbb{E}}[u'(c')] - \mathbb{E}[u'(c')] \;\approx\; \tfrac{1}{2}\,(\hat{\sigma}_\psi^2 - \sigma_{\psi,g}^2)\,\mathbb{E}\big[u'''(c')\,c_y^2 + u''(c')\,c_{yy}\big], \tag{11}$$

where c_y and c_{yy} are the first two income-derivatives of the consumption policy. Under the CRRA used here ($\gamma = 2$, so $u''' = \gamma(\gamma+1)c^{-\gamma-2} > 0$) the leading term is positive, so an agent who perceives less permanent risk than she faces under-saves for precaution and opens the wealth and marginal-propensity-to-consume wedge that I will quantify in Section 10. The operative gap is the permanent one: a permanent surprise accumulates in the random walk and shifts the whole future consumption path, whereas a transitory belief error is buffered within the period and barely reaches c_y.

The incidence of this channel is gated by liquidity. Because perceived risk acts only through precautionary saving, the saving response to the permanent-variance belief, $\partial a'/\partial\hat{\sigma}^2_\psi$, is zero at the borrowing constraint $a' = \underline{a}$ and strictly positive only where the household holds slack. This acts in the opposite direction to a level belief error, that is, a misperception of expected income rather than its risk, which reaches a constrained, high-MPC household one for one. In the hard-constraint limit the two incidences are disjoint: a variance error leaves the constrained household untouched and moves only the unconstrained one, while a level error would do the reverse. In the cross-section, though, the ordering of the wedge follows the belief gap and not the constraint. The gap grades with legibility, small at the bottom, which labels little, and largest at the top. In the model of Section 10 the wealth wedge per unit of gap is nearly flat across groups, so liquidity reinforces the incidence only at the margin. The bottom therefore holds the most distorted transitory prior of any group and still bears the smallest behavioural wedge: its distortion sits in the prior, buffered within the period, while its permanent belief is essentially unbiased.

Level shocks versus risk shocks

The wedge to this point is a stationary object, and how long a shock moves it depends on whether the shock is a one-off surprise or a lasting rise in risk. The permanent belief is a state driven by the squared surprises of the rule (2): a single surprise raises it by $\mu\lambda(\rho)$ per unit of the squared change, the effect decays geometrically at rate $1-\mu$, to $\mu\lambda(\rho)(1-\mu)^h$ after h periods, and the saving wedge inherits this path. A level shock, the kind a recession delivers, is a single such surprise, so it only traces this transient path: the belief jumps on impact and the constant gain discounts it away within a quarter and leaves no lasting wedge.

A risk shock, a persistent rise Δ in the true permanent variance, keeps the squared surprises large for as long as it lasts, and under the constant gain the agent books only a fraction of the rise as permanent, so her perceived permanent risk climbs but settles short of the truth for as long as the risk stays high. How much she passes through depends on how well she reads her pay. The recursion runs on her own noisy reading, so the reading noise that inflates the bottom's transitory prior also damps its pass-through, and the bottom's perceived risk rises least: the prior that cancels from the resting belief re-enters in the response to a shock, as Appendix D.1 details.[8] The behavioural wedge nonetheless lands at the top, for the same reason as in the steady state: the saving-relevant permanent gap grades with legibility and is largest there, liquidity matters only at the margin, and the bottom, which passes the least of the rise through, has the least to act on. Section 9 embeds the measured beliefs in the quantitative model, and Section 11 takes both shocks to it.

9 A heterogeneous-agent model

To trace what those perceptions imply for behaviour, I embed the estimated belief structure in a closed general-equilibrium incomplete-market economy and compare each income group under

[8]The shortfall separates into two parts: the under-reaction an unbiased agent with the same noisy reading would carry, largest at the bottom, and the legibility excess the mechanism adds on top, $(\lambda^*_g - \lambda(\rho))\Delta$ in the closed-form accounting, largest at the top.

its estimated beliefs with an otherwise identical full-information benchmark. The gap between the two regimes, in precautionary wealth and the marginal propensity to consume, is the belief wedge.

Setup: a general-equilibrium incomplete-market economy

Demographics and preferences. The economy is a monthly overlapping-generations life-cycle buffer-stock model following Wang (2023), closed with a single-asset technology. A worker enters at age 25, works for forty years, retires at 65 onto a pension replacing sixty percent of her permanent income after a one-fifth step-down in the permanent wage at the boundary, and dies with certainty at 85, with a small constant mortality risk before then. Each death is matched by a new entrant drawn across the income distribution, so the cross-section is stationary through the finiteness of life. Preferences are CRRA with $\gamma = 2$, and at the final age she values the assets she leaves directly rather than spending them to zero.

Income. Income follows the permanent-transitory process of Section 4 scaled by a deterministic age-wage profile, $y_{i,t} = \exp(p_{i,t})\, G_\tau\, \xi_{i,t}$. Here $p_{i,t} = p_{i,t-1} + \psi_{i,t}$ is the permanent random walk and $\xi_{i,t}$ the transitory state, with their variances taken from Table 2, while G_τ is the hump-shaped age profile Wang (2023) uses to separate predictable life-cycle earnings growth from risk. When employed the worker draws $\xi = \exp(\theta_{i,t})$; when unemployed she receives a benefit replacing half her permanent income, and in retirement she receives the pension.

Mobility. The permanent innovation arrives while she is employed, consistent with the same-job stayer panel its variance is measured on, and it is the one she learns about in Section 5. Accumulated over a career, the innovations carry her across the income distribution: a run of positive permanent raises lifts a worker out of the bottom group without any change of job, and the deterministic profile G_τ adds a predictable component to the same drift. A current-income tercile therefore mixes types, a distinction the results return to. The anchors themselves are household-income groups, which the model reads as positions in individual earning power. Appendix C.6 checks the realised side of the mapping by re-sorting on own earnings.

Unemployment. Caplin et al. (2026) find that the possibility of quitting or being fired is one of the key drivers of subjective earnings risk, so the transition margin cannot be left out of a model of perceived risk. The SCE question conditions it away by fixing the job, which is what makes the wage-risk measurement clean, so following Wang (2023) the budget restores it through a two-state employment block, with rates taken from the SCE's own perceived job-loss and job-finding probabilities; a second finding of Caplin et al. (2026), that average subjective separation probabilities and expected durations track administrative records even by age, is what allows perceptions to supply them.[9] The rates come to monthly separation 0.015/0.011/0.011 and job-finding 0.23/0.27/0.29 from bottom to top, a steady-state unemployment rate of 6.2/4.0/3.7 percent, higher and more persistent at the bottom and a little below the seven-percent flow benchmark of Shimer (2005), with replacement income half of permanent. The transition margin is therefore

[9] For each employed respondent I take the SCE's reported percent chance of losing the main job over the next twelve months and, conditional on losing it, of finding a new one within three months, average each within income group with survey weights, and convert a horizon-H probability P to a monthly hazard $-\log(1-P)/H$ (with $H = 12$ for separation and $H = 3$ for finding), the conversion Wang (2023) applies to the same SCE questions. The tercile-mean twelve-month job-loss probabilities are 16.9/12.7/12.2 percent and the three-month finding probabilities 50.2/56.0/57.6 percent.

perceived accurately where wage risk is not, and because its gradient is common to both regimes it shapes the wealth distribution, the hand-to-mouth share, and the constraint incidence the wedge acts through without being a source of the wedge, as Wang (2023) emphasises. Beliefs update only while employed and freeze during unemployment.

Beliefs. The labelled share, the update gain, and the perceived transitory prior are functions of current income, $\rho(y)$, $\mu(y)$ and $\sigma^2_{\theta,\mathrm{perc}}(y)$, interpolated through the tercile anchors of Sections 6 and 7, and a worker slides along them as her income drifts.

Normalisation. Following Carroll (2006) I normalise by permanent income: the state is cash-on-hand per unit of permanent income, the per-period draw is the transitory $\exp(\theta)$, and permanent shocks enter through the growth factor $G_\tau \exp(\psi)$. The problem is scale-invariant, so I report each group's mean monthly SCE income (\$2,415 / \$6,014 / \$13,604) only to fix magnitudes.

Technology and market clearing. In general equilibrium I close the economy with a constant-returns technology, $Y = ZK^\alpha N^{1-\alpha}$, capital share $\alpha = 0.36$ and monthly depreciation from an annual 2.5 percent, the single asset carrying both the household buffer and the return. The unemployment benefit and the pension are financed by flat balanced-budget taxes on labour income, so disposable income is net of a payroll and a social-security rate. The capital market clears against household asset supply, and I pin the productivity normaliser Z once from the FIRE economy at a target rate of three percent, holding it fixed across regimes so the FIRE-versus-belief rate gap is the general-equilibrium wedge.

Two regimes. I solve the model twice. Under the *belief* regime the agent perceives wage risk through the estimated rule and her belief evolves; under *FIRE* she knows the true variances. Everything else is shared, so any gap is the belief wedge. FIRE here means correct wage-risk beliefs.

The household problem

$$V_\tau(m, p, s, \hat\sigma^2_\psi) = \max_{0 \le c \le m} \left\{ u(c) + \beta\,(1 - D)\,\widetilde{\mathbb{E}}\left[\Gamma'^{1-\gamma}\, V_{\tau+1}(m', p', s', \hat\sigma^2_{\psi'})\right] \right\},$$
$$m' = \frac{R}{\Gamma'}\,(m - c) + \xi', \qquad \Gamma' = G_\tau \exp(\psi').$$

where m is normalised cash-on-hand, τ age, p the income node, s employment status, D monthly mortality, $R = 1 + r$ the gross return, and $\widetilde{\mathbb{E}}$ the expectation under the agent's *perceived* process. The terminal age carries the bequest value in place of a continuation value. The budget uses *realised* income while expectations are taken under the perceived process, so beliefs act only through the saving motive (Section 8).

The belief follows the constant-gain rule of Section 5, $\hat\sigma^2_{\psi,t} = (1 - \mu(y))\hat\sigma^2_{\psi,t-1} + \mu(y)\,\tilde\lambda(y)\,(\Delta\tilde y_t)^2$, on the agent's own netted reading $\tilde y$ of (5), with the gain scaled so that the belief rests at the value (8) implies (the scaling is in Appendix D.1). Table 4 collects each input. The steady-state comparison uses this resting value, which adds no state dimension. The dynamic state is carried explicitly for the risk-shock response.

Stationary equilibrium. For each regime a stationary equilibrium is a family of age-indexed value functions, policies $c_\tau(m, p, s, \hat\sigma^2_\psi)$, and an invariant distribution Φ such that the policies solve the household problem, Φ is invariant under them, the capital market clears at the equilibrium interest rate r, and the government budget balances. The invariance runs over the income process, the demographics, the employment transitions, and the belief law of motion, which is degenerate at the truth under FIRE. I read moments of Φ off a simulated panel of $N = 30{,}000$ households over 1,200 months (720 burn-in), with the belief and FIRE branches driven by a single common shock path, so each worker sits in the same income tercile under both regimes and the belief-minus-FIRE gap is a within-worker difference free of sampling noise.

Calibration

The belief block is carried as Sections 6 and 7 deliver it, the gain and the transitory prior estimated, the labelled share pinned inside its measured band, and the attribution and the resting permanent belief implied in closed form by (7) and (8), within half a point of their measured-share values. The calibration fits a single remaining parameter, a discount factor common across agents and matched to average liquid wealth; everything else is assigned or carried as measured, and Table 4 collects the values. The model is block-recursive: beliefs evolve on the realised income process and never on assets or consumption, so the belief block is measured and estimated independently of preferences.

Parameter	Symbol	Value (Bot / Mid / Top)	Source or target
A. Preferences and assets (common)			
Risk aversion	γ	2	Assigned (standard)
Annual interest rate	r	2%	Fixed for PE; clears near 3% in GE
Borrowing limit	$\underline{a}$	0	Assigned (standard)
Discount factor	β	0.917	Calibrated to mean liquid wealth ($\approx$ 1.8 mo)
B. Employment block (heterogeneous by income; common across regimes)			
Separation rate	s_g	0.015/0.011/0.011/mo	SCE perceived, monthly hazard
Job-finding rate	f_g	0.23/0.27/0.29/mo	SCE perceived, monthly hazard
Replacement rate	b	0.5	Standard convention
Pension replacement	ς	0.6	As in Wang (2023)
C. Realised income process by group, $(\%)^2$			
Permanent variance	$\sigma^2_{\psi,g}$	11.9/11.6/11.2	SIPP (Table 2)
Transitory variance	$\sigma^2_{\theta,g}$	2.5/4.4/5.8	SIPP (Table 2)
Mean monthly income	–	\$2,415/6,014/13,604	SCE (scale anchor only)
D. Beliefs by group: the labelled share pinned, the rest implied or carried			
Labelled share	ρ_g	0.119/0.197/0.369	Pinned in measured band (Section 7)
Update gain	μ_g	0.629/0.910/0.950	GMM estimate, carried
Attribution	λ_g	0.677/0.519/0.398	Implied from ρ_g, eq. (7)
Transitory prior	$\sigma^2_{\theta,\mathrm{perc},g}$	9.65/3.55/4.12	GMM estimate, floor-censored (Mid binds)
Resting permanent belief	$\hat{\sigma}^2_{\psi}$	11.47/10.61/9.11	Implied, eq. (8)

Table 4: Model calibration. The employment block is heterogeneous by income but common to both regimes; under FIRE the belief block of panel D is held at the truth.

Preferences, a single discount factor to average liquid wealth. I set the discount factor so that the FIRE economy, at the fixed rate of 2 percent annual, matches the average liquid wealth in the data, about 1.8 months of permanent income, which gives an annual β of 0.917.[10] A single calibrated β pins the *level* of wealth and of the marginal propensity while leaving the cross-section of both an outcome of risk. The alternative of a consensus discount factor, as in Wang (2023), lets the economy over-accumulate, and an income-varying $\beta(y)$ would build the cross-sectional gradient into preferences.[11]

10 The belief wedge

I read the wedge in two settings: the cross-section, with prices fixed at the calibrated rate so the gap between the two regimes is the belief channel alone, and the closed economy, where the interest rate clears and adds the price response the fixed-rate reading shuts down.

The cross-sectional wedge

What the measured beliefs imply. Table 12 (Appendix F) sets the stationary perceived variance against FIRE in the same SIPP units, so the comparison is internal to the model and free of the level gap between the two surveys. The model's perceived variances accordingly sit seven to thirteen times above the reported levels of Section 3, the convention stated with the identity in Section 5: the reported shortfall lives in the compounded permanent term and is read as an upper bound, the intercept crosses the surveys unrescaled, and carrying the reported level itself would hand every group an under-perception so large that no group difference could matter. Perceived twelve-month risk falls with income, from 153 through 139 to 124 against a FIRE benchmark flat near 148, a ratio of 1.03/0.94/0.84: the bottom perceives about three percent more total risk than it faces, its inflated transitory prior outweighing a permanent discount of about four percent, while the middle and the top perceive six and sixteen percent less, the labelling discount deepening with income faster than the prior falls, as Section 8 derives.

Figure 7 compares the belief regime with an otherwise identical FIRE benchmark at the calibrated interest rate, and Table 13 (Appendix F) reports the numbers. With the discount factor matched to average liquid wealth, the FIRE economy holds 2.3/1.7/1.5 months of permanent income, with twelve-month MPCs of 0.47/0.41/0.38 and hand-to-mouth shares of 43/41/42 percent, the propensities in the data's range and the shares a little above it. Mean wealth in months falls mildly with income where the median in the data rises, a miss that survives the workers-only split of Table 13.[12] The belief wedge is computed with this calibration held fixed.

[10] Targets from the SCE Household Finance module, the liquid-wealth concept of Kaplan et al. (2014): median liquid wealth about 0.5/2.0/3.0 months across the three groups, averaging near 1.8.

[11] Calibrating on the FIRE regime is a normalisation: under the measured beliefs average wealth falls short of the FIRE target by a few percent, so recalibrating on the belief regime would leave the wedge essentially unchanged. The resulting β sits below the 0.95–0.96 of the life-cycle literature (Gourinchas and Parker, 2002), the price of holding a one-asset economy to 1.8 months of *liquid* wealth rather than to total net worth. I solve by the endogenous grid method (Carroll, 2006).

[12] The direction is the one-asset buffer at work: wealth in months is a buffer against risk, and the risk that matters most, persistent unemployment, falls with income, and the rising liquid-wealth profile in the data reflects the life-cycle, illiquid, and bequest saving that a single asset cannot separate from the buffer. Income-varying patience would reproduce that profile; the common discount factor forgoes it to keep the cross-section an outcome of risk.

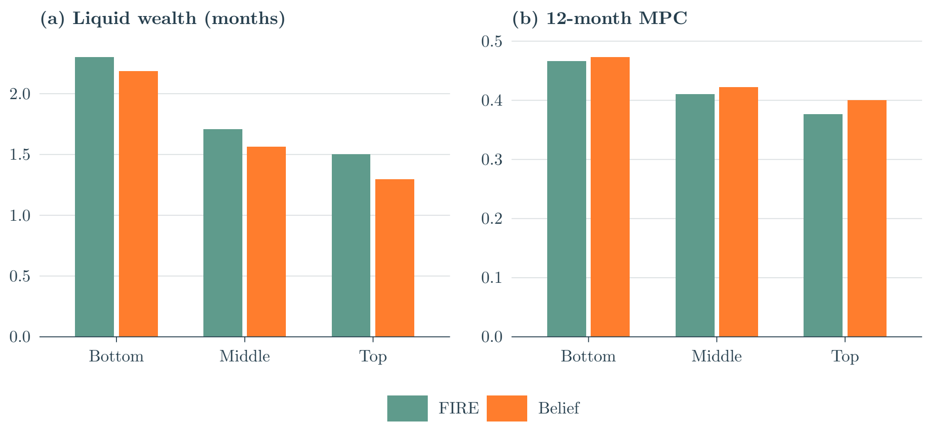

Figure 7: Mean liquid wealth (months of permanent income) and the 12-month MPC by income group, FIRE versus the belief regime (Table 13).

The wedge in wealth rises monotonically with income: the belief economy holds 5.1, 8.3, and 13.8 percent less than FIRE from bottom to top under the reconciled shares, and 7.0, 10.9, and 17.6 percent less at the detrended upper edge of the measurement band (Appendix D.2). The wedge in the marginal propensity to consume concentrates at the top, +0.7, +1.3, and +2.4 points, and +3.1 points at the upper edge. Both profiles track the permanent-belief gap, smallest at the bottom and largest at the top, and the wedge per unit of belief gap is nearly flat across groups, so liquidity adds little.

The wedge is robust to the rate it is read at: recalibrating the discount factor to the same liquid-wealth target at the general-equilibrium rate of three percent leaves it at 5.1, 8.3, and 13.6 percent, so the wealth calibration pins the cross-section at either rate (holding the discount factor fixed instead, the rate alone would widen it to 6.8, 10.5, and 16.4 percent). A decomposition attributes it entirely to the permanent belief: solving the model with the permanent discount alone, the transitory prior held at the truth, reproduces the full wedge on both margins, while the complementary regime, the carried prior with the permanent belief at the truth, moves neither wealth nor the marginal propensity measurably. The prior is behaviourally inert, and the permanent-belief discount accounts for the whole wedge.

The bottom is least moved on either margin, +0.7 points on the MPC and 5.1 percent on wealth, much as Section 8 anticipates: a bottom-income type labels little, so its permanent belief is discounted by under four percent and carries the least saving-relevant distortion. Composition does part of the rest: just over half of the bottom current-income tercile is retirees, against roughly a quarter and a tenth above, and among its workers the bottom is the least constrained of the three groups, with hand-to-mouth shares of 27/38/38 percent from bottom to top. The smaller wedge is therefore a fact about beliefs and composition, with the borrowing constraint playing little part, and restricting to workers and re-terciling among them sharpens the shape mildly, to −5.8/−9.7/−15.1 percent on wealth and +0.9/+1.5/+2.7 points on the marginal propensity. The middle is moderately distorted on both margins, −8.3 percent on wealth and +1.3 points on the MPC: unconstrained and labelling more of its transitory pay, it under-saves. The top responds most on both, holding 13.8 percent less wealth and consuming over two points more of a windfall over the year: it labels its transitory pay away most completely, under-perceives the permanent risk that drives saving most, and so holds the most-discounted permanent belief and the largest wedge. The permanent belief lies below the truth at all times, so the wedge is a property of the stationary state, in place before any shock arrives.

The size of the wedge turns on the reading of the updating rule, with sampling error second order. Under the Bayesian reading, whose agent raises her attribution as netting cleans her signal, the permanent belief is unbiased and the wedge is zero by construction, so the quantitative results stand or fall with the constant attribution and the evidence of Section 8 that favours it. Under the constant attribution at the measured shares, the wedge is the baseline of 5.1, 8.3, and 13.8 percent of wealth; under the belief-anchored variant of Appendix E.2, which takes the estimated attribution responses at face value, it roughly doubles. The shape, rising with income and concentrated at the top, is common to both, and I read the baseline as the estimate and the variant as the outer edge, because only the baseline rests on the measured delivery of pay.

The closed economy

Closing the economy lets the saving wedge move the interest rate. At the calibrated discount factor the FIRE economy clears at the three-percent target by construction, with the productivity normaliser held fixed across regimes (Section 9), and the belief economy, which under-perceives permanent risk, supplies about five percent less aggregate capital and clears at a rate 0.21 percentage points higher, 3.2 against 3.0 percent (Table 14, Appendix F). The response corroborates Wang (2023) in general equilibrium: lower perceived risk lowers precautionary supply and raises the equilibrium return. Under the measured shares the movement is modest, however, and the wedge shows in the cross-section more than in the aggregates. The magnitude is the one-asset response, in which a single liquid asset carries both the buffer and the return. A credible quantification of how the wedge moves marginal propensities and the transmission of aggregate shocks needs a two-asset closure that separates liquid from illiquid wealth (Kaplan and Violante, 2022).

11 Dynamics

The belief is a state, so the wedge moves as the belief moves. It responds to the two primitives the belief rests on: the legibility of pay, the share ρ a worker can label and net out, and the permanent risk the pay carries, σ_ψ^2. I trace a change in each through the belief channel with prices held fixed, the dynamic-belief economy of Section 9 against an otherwise identical full-information economy on the same realised path. The delivery change is the mechanism's own experiment and comes first, the risk change is the aggregate experiment of the literature read through the mechanism, and a recession, which changes neither primitive, closes the section as a check.

Figure 8: Response to a legibility reform that raises the labelled share of the bottom and the middle to the top's level of $\rho = 0.37$, belief regime, reform minus no-reform, by income group. Perceived risk is the twelve-month perceived variance in $(\%)^2$, and consumption and liquid assets are in percent of the no-reform path. Paths average twelve simulated cohorts, and the perceived-risk panel is a three-month moving average of the belief-state mean.

A legibility shock. The experiment the mechanism most directly invites changes how legibly pay is delivered and leaves the risk itself untouched. Let a reform raise the labelled share ρ of a low-paid worker to the level a salaried worker already holds, so that variable shift counts and cash supplements arrive as a single itemised figure while the true income process is unchanged; the full-information economy then does not move, and the entire response is the belief channel, shown in Figure 8. As the cleaner signal arrives, the worker nets more of her pay before she updates and reads what remains as less telling about her permanent earning power, and her perceived permanent risk falls within a quarter, sharply for the newly legible bottom and middle and barely for the top. She then saves less against a risk she now under-perceives: consumption rises on impact and the precautionary buffer erodes over one to two years, by 2.9 and 3.0 percent for the bottom and the middle against 1.1 for the top, whose smaller move arrives through the treated workers who drift into it. The belief adjusts within months and the buffer over years, the two speeds the figure separates.

The incidence follows which workers the reform reaches. Formalising low-paid work moves the bottom and the middle towards the under-perception the top already carries, so the response lands on them. A broad rise in legibility across all pay would instead fall hardest on the top, where the labelled part scales with the largest transitory variance. Either way the direction is the same: making pay more legible lowers precautionary saving, because a cleaner permanent signal, read under an attribution that does not re-scale to it, registers as less risk. The response is accordingly conditional on the constant attribution, since the Bayesian of Section 8, re-scaling as netting cleans her signal, would leave her saving unmoved. The value of the exercise is the design rather than the magnitude: the steady-state wedge lives in a belief level no survey observes, while the response to a delivery reform is a behaviour a payroll change with saving data could confirm or refute.

Figure 9: Response to a persistent 40% rise in the true permanent wage variance, belief regime minus FIRE, by income group. Perceived risk is the gap in the rise in twelve-month variance in $(\%)^2$, and consumption and liquid assets are in percent of the no-shock baseline. Paths average twenty-four simulated cohorts, and the perceived-risk panel is a three-month moving average of the belief-state mean.

A risk shock. The mechanism also governs how the distribution absorbs a rise in the risk itself. Let the true permanent variance rise persistently, by forty percent, borrowing the magnitude from the literature (Bayer et al., 2019). Full information recognises the higher risk at once and rebuilds its buffer, holding fifteen to nineteen percent more liquid wealth within two years. The belief regime books only a fraction of the now-larger squared surprises as permanent, 0.37/0.57/0.47 across the groups, smallest at the bottom, where the noisy reading damps it, and largest at the middle, where neither the reading noise nor the netting is strong, so its perceived permanent risk rises far less and it accumulates 14.7/18.1/18.7 percent less additional liquid wealth than full information over the two years, each regime read against its own no-shock path (Figure 9). The shortfall is the same under-reaction the legibility reform exploits, now against a moving truth rather than a moving signal, and the precautionary response to rising risk is muted most at the top, where legibility is highest. A recession, by contrast, a one-off three-percent fall in the level of income, changes neither primitive: its single surprise lifts the belief only transiently, as Section 8 anticipates, and the two regimes leave the episode with the same buffer. The wedge lives in the primitives the belief rests on, how pay is delivered and how risky it is, rather than in the business cycle.

12 Conclusion

This paper set the risk workers perceive against the risk their wages carry, on the same jobs, and found that the assumption equating the two fails systematically across the income distribution. Workers at the bottom perceive about double the wage risk of workers at the top, hold that belief more persistently, and revise it more strongly after a realised wage shock, while the risk itself, if anything, rises mildly with income. A constant-gain rule estimated within person organises all three gradients and locates the deviation on both sides of the truth: the bottom reports transitory risk far above its realised counterpart, the middle and the top under-perceive their permanent risk, and reported totals lie below the full-information benchmark in every group.

I propose that the gradient reflects how pay is delivered. The share of pay movement that arrives labelled on the paycheck rises from about an eighth at the bottom to over a third at the top, and a worker with legible pay nets the labelled part out before she updates: her transitory prior falls to the unlabelled remainder, her belief moves faster, and she books less of each surprise as permanent. Measured from the itemised pay components of the same records, the share predicts where the middle and top priors settle, to within a standard error, the ordering of belief persistence, and about half of the estimated gradient in the response to news; the bottom's prior stands four standard errors above the level its share implies, an excess the mechanism attributes to the unresolved reading of fragmented pay.

In the life-cycle economy the deviation becomes a tilt in the cross-section: a calibration that equates perceived with realised risk overstates precautionary wealth and understates the propensity to consume by amounts graded in income. The error is negligible at the bottom, whose permanent belief is nearly unbiased and whose inflated transitory prior the buffer absorbs, and largest at the top, which under the measured beliefs holds fourteen percent less liquid wealth and consumes over two points more of a windfall within the year than under full information. The spread of propensities between the bottom and the top narrows by about a sixth, so an exercise whose shock or policy falls on higher incomes under-predicts the consumption response of the households the calibration assigns the lowest propensities; the aggregates move little, the capital stock by five percent and the interest rate by about twenty basis points, so the correction is distributional. Where the question involves shocks to risk itself, the calibration also overstates the precautionary response to them, since agents book only part of a rise in risk as permanent and rebuild fifteen to nineteen percent less buffer than full information within two years.

The recommendation for the practitioner is then direct: keep the realised process in the budget, and hand the agent a permanent variance discounted below it by an income-graded factor, measurable from the labelled share of pay in payroll records. The transitory side stays at the truth, and the discount adds no state variable, so the whole correction enters a standard calibration as a change of parameters.

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A 12-month FIRE variance

Rolling $e_t = p_t + \theta_t$ forward H months, the permanent piece accumulates one fresh innovation per month, $\mathrm{Var}(p_{t+H} - p_t) = H\sigma^2_{\psi,g}$, while the iid transitory piece contributes only its endpoints, $\mathrm{Var}(\theta_{t+H} - \theta_t) = 2\sigma^2_{\theta,g}$. The shocks are independent, so $\mathrm{Var}(e_{t+H} - e_t) = H\sigma^2_{\psi,g} + 2\sigma^2_{\theta,g}$, which at $H = 12$ is (1).

Group	FIRE benchmark ($12\sigma^2_{\psi,g} + 2\sigma^2_{\theta,g}$)	SCE observed $\bar{\sigma}^2_{\text{perc}}$	ratio
Bot	147.7	20.6	7.2×
Mid	148.0	11.3	13.1×
Top	146.3	11.4	12.8×

Table 5: FIRE-implied 12-month variance versus SCE-reported perceived variance, in $(\%)^2$.

Appendix C.7 recomputes the benchmark under the same-hours restriction, where it falls to 79, 85, and 98 $(\%)^2$ and the ratios to 3.8×, 7.5×, and 8.6×.

B Closed form for λ_g under partial labelling

This appendix derives the closed form (7): the attribution on the full pay change is the unbiased attribution on the unlabelled residual, attenuated by the labelled pay that rides in the change. Split the transitory shock $\theta_t = \theta^l_t + \theta^u_t$ into independent pieces with variances $\rho_g\sigma^2_{\theta,g}$ and $(1-\rho_g)\sigma^2_{\theta,g}$. Then $\Delta e_t = u_t + l_t$, with unlabelled residual $u_t = \psi_t + \theta^u_t - \theta^u_{t-1}$ and labelled component $l_t = \theta^l_t - \theta^l_{t-1}$, so $V_{u,g} = \sigma^2_{\psi,g} + 2(1-\rho_g)\sigma^2_{\theta,g}$, $V_{l,g} = 2\rho_g\sigma^2_{\theta,g}$, and $V_{u,g} + V_{l,g} = \sigma^2_{\psi,g} + 2\sigma^2_{\theta,g} = \bar{S}_{m,g}$, the variance of the full monthly change.

The rule of Section 5 updates on the full observed change $(u+l)^2$, so the attribution recovered by (4) is the unbiased residual attribution $\sigma^2_{\psi,g}/V_{u,g}$ attenuated by the projection of u^2 on $(u+l)^2$, $\lambda_g = (\sigma^2_{\psi,g}/V_{u,g})\,\mathrm{Cov}(u^2,(u+l)^2)/\mathrm{Var}((u+l)^2)$. Under joint Gaussianity with $u \perp l$, $\mathrm{Cov}(u^2,(u+l)^2) = 2V^2_{u,g}$ and $\mathrm{Var}((u+l)^2) = 2(V_{u,g}+V_{l,g})^2$, so

$$\lambda_g \;=\; \frac{\sigma^2_{\psi,g}V_{u,g}}{(V_{u,g}+V_{l,g})^2} \;=\; \frac{\sigma^2_{\psi,g}\left(\sigma^2_{\psi,g} + 2(1-\rho_g)\sigma^2_{\theta,g}\right)}{(\sigma^2_{\psi,g} + 2\sigma^2_{\theta,g})^2},$$

which is (7).

This closed form is one of three readings of the estimating equation, and the scorecard's ratio comparison does not turn on the choice among them. Under the netting reading, in which the agent nets first and updates on u_t^2 at the fixed attribution $\lambda^*_g = \sigma^2_{\psi,g}/\bar{S}_{m,g}$, the coefficient in (4) is the projection $\lambda^*_g\,(V_{u,g}/\bar{S}_{m,g})^2 = \sigma^2_{\psi,g}V^2_{u,g}/\bar{S}^3_{m,g}$. The predicted bottom-to-top and middle-to-top ratios become 1.99 and 1.48 in place of 1.69 and 1.31, and both pairs lie within one standard error of the estimated ratios (Appendix D.3). The implemented form, updating on the agent's own noisy netted reading with the rescaled attribution $\tilde{\lambda}$ (Appendix D.1), shares its resting mean with both, so the readings differ in the coefficient and not in the resting belief.

The Gaussian fourth-moment identities are checked directly on the component panel of Appendix D.2: replacing them with the sample projection $\mathrm{Cov}(u^2, (u+l)^2)/\mathrm{Var}((u+l)^2)$ moves the predicted attribution ratios from 1.69 and 1.31 to 1.54 and 1.13 under the baseline trim, still about one standard error from the estimated ratios, while the untrimmed projection is dominated by the labelled bonus tail documented there.

C Belief estimates and robustness

Table 6 reports the headline joint-GMM estimates of (4) plotted in Figure 4, C.1–C.6 check their robustness, C.7 constructs the same-hours counterpart of the realised process, and C.8 measures the twelve-month benchmark directly.

Group	$\sigma^2_{\theta,g}$ (SIPP truth)	$\sigma^2_{\theta,\mathrm{perc},g}$ (GMM)	λ_g (GMM)	μ_g (GMM)	Hansen J p	N
Bot	2.52	**9.65** (1.76)	+**0.079** (0.031)	0.629 (0.055)	0.23	1,531
Mid	4.42	2.96 (0.94)	+0.046 (0.015)	0.910 (0.089)	0.23	7,501
Top	5.83	4.12 (0.82)	+0.026 (0.013)	0.950 (0.060)	0.23	11,331
Wald ($B = T$)	—	$p =$ **0.004**	$p = 0.12$	$p =$ **0.0001**	—	—

Table 6: Joint Blundell-Bond system GMM on (4), by income group (the estimates of Figure 4). Two-way clustered SEs (respondent, quarter) in parentheses. Instruments: $\bar{\sigma}^2_{\mathrm{perc}}$ at lags 3–4, the shock at lags 1–3, plus BB level moments, collapsed. N counts observations, and respondent counts are 269/1,050/1,452. The Hansen J is a single joint-system statistic repeated across rows. Bold p-values mark Wald rejections of cross-group equality at 5%, and bold estimates mark the bottom's prior and attribution (Section 6).

C.1 Occupation × time fixed effect

Table 7 adds an occupation × year-month FE to the shock construction: the persistence and attribution gradients survive, attribution sharpens, and the prior gradient keeps its sign but loses significance.

	Canonical			Occupation × time FE		
Group	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}^2_{\theta,\mathrm{perc},g}$	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}^2_{\theta,\mathrm{perc},g}$
Bot	0.629 (0.055)	+**0.079** (0.031)	**9.65** (1.76)	0.649 (0.061)	+**0.088** (0.039)	**7.30** (2.66)
Mid	0.910 (0.089)	+0.046 (0.015)	2.96 (0.94)	0.923 (0.090)	+0.037 (0.010)	2.62 (0.98)
Top	0.950 (0.060)	+0.026 (0.013)	4.12 (0.82)	0.959 (0.060)	+0.017 (0.008)	4.19 (0.73)
Wald μ (T−B)		$p =$ **0.0001**			$p =$ **0.0003**	
Wald λ (B−T)		$p = 0.116$			$p = 0.068$	
Wald σ^2 (B−T)		$p =$ **0.004**			$p = 0.260$	

Table 7: Canonical versus occupation × year-month FE in the wage-shock construction. Standard errors in parentheses.

C.2 Lag depth, trim, COVID, and the bottom's trend

Specification variants. Table 8 varies the lags, trim, and COVID handling one at a time: the Hansen J never rejects, attribution stays monotone in every row, the persistence Wald survives all but the tighter trim, the prior Wald all but the COVID exclusion.

	$\hat{\mu}_g$			$\hat{\lambda}_g$			$\hat{\sigma}^2_{\theta,\text{perc},g}$			Wald p (B vs T)		
Variant	B	M	T	B	M	T	B	M	T	μ	λ	σ^2
Canonical	0.629	0.910	0.950	+0.079	+0.046	+0.026	9.65	2.96	4.12	**0.0001**	0.116	**0.004**
Trim 2.5/97.5	0.824	0.846	0.891	+0.096	+0.055	+0.014	11.16	4.05	4.82	0.067	0.188	**0.000**
$\bar{\sigma}^2$ lags 2–3	0.649	0.951	0.876	+0.077	+0.050	+0.020	9.85	2.87	4.44	**0.000**	0.080	**0.004**
$\bar{\sigma}^2$ lags 4–5	0.590	0.899	0.948	+0.104	+0.046	+0.021	8.94	3.01	4.38	**0.000**	0.063	**0.040**
Shock lags 0–1	0.681	0.936	0.940	+0.084	+0.016	+0.012	9.67	4.42	4.89	**0.013**	0.061	**0.012**
Shock lags 2–4	0.672	0.908	0.948	+0.098	+0.040	+0.027	8.67	3.21	4.10	**0.002**	**0.010**	**0.016**
Exclude 2020–21	0.642	0.992	0.886	+0.140	+0.043	+0.013	6.40	3.13	4.88	**0.021**	0.052	0.637

Table 8: One-at-a-time variations around the canonical specification. Bold p-values reject cross-group equality at the 5% level.

The bottom's trend. The bottom series drifts upwards (+0.83 per year, $t = 3.7$), which strains the mean stationarity behind the Blundell-Bond level moments, though the belief block does not rest on them. A linear detrend or calendar-year effects leave the bottom prior at 9.74 and 9.57 against the canonical 9.65, the bottom gain at 0.62 and 0.64 against 0.63, the gain Wald below 0.0001, and the prior Wald at most 0.008.

The difference-in-Hansen statistic on the level moments does not reject: $C = 1.20$ and 1.63 on two degrees of freedom at the bottom and top ($p = 0.55$ and 0.44), and the middle's statistic is negative. Dropping the level moments preserves the point estimates, with the bottom prior at 9.22 (1.81), but the gain loses its precision.

C.3 The sign of news

Positive wage news carries roughly twice the belief weight of negative news, an asymmetry I treat as suggestive rather than established. Pooling the groups for power and splitting the shock by sign, both components load positively, with $\lambda^+ = 0.055$ versus $\lambda^- = 0.027$, significant across cluster choices ($N = 20{,}363$, Hansen J $p = 0.11$). The direction is the opposite of what loss aversion would predict (Kahneman and Tversky, 1979): upside news is read as informative about future earning power, while downside news is more often dismissed as transitory. Two caveats keep the reading suggestive: the extra weight may reflect the size of the surprise rather than its sign, since expected wage growth was positive over the sample, and pooling leaves open whether the asymmetry itself differs by income.

C.4 Numeracy and inflation uncertainty

The level gradient of Section 3 survives conditioning on numeracy and on inflation uncertainty. The SCE's numeracy module classifies 47.0 percent of bottom-group respondents as low-numeracy, against 28.4 percent in the middle and 11.1 at the top. On the full matched panel, a broader sample than the estimation panel, the bottom-to-top difference in perceived variance is 10.6 (1.0); it falls to 2.4 (0.8, $p = 0.002$) among high-numeracy respondents and rises to 17.0 (2.1) among low-numeracy ones.

Inflation uncertainty is itself steeply graded, 27.9/15.2/8.5 across the groups, and controlling for it absorbs about seventy percent of the bottom-to-top wage-risk gap, leaving a wage-specific gradient of +3.09 (0.60). On the over-control reading of Section 3, both controls remove part of the reading-noise channel itself and the conditioned gaps are lower bounds; read instead as mechanical reporting width, they are the cleaner estimates. The gradient survives either reading, its size bracketed by the two.

Re-estimating the belief block (4) within high-numeracy respondents reproduces the middle and the top, with priors of 2.35 (0.85) and 4.93 (1.02), at or below their realised values as in the baseline. The bottom retains only 75 estimation observations under the numeracy-module overlap, too few for the system GMM, so its prior excess cannot be checked at the estimation level.

Three checks bound the part reporting mechanics can carry. First, bottom respondents use 5.7 of the ten histogram bins, against 4.9 in the middle and 4.6 at the top, and restricting to reports that use three or more bins leaves the gradient at 23.0/12.3/12.2. Second, low numeracy widens the reported inflation density by more than the wage density in every group, with wage-against-inflation log-variance gaps of 0.92 against 1.17 at the bottom, 0.47 against 0.76 in the middle, and 0.15 against 0.62 at the top: the widening is a generic reporting trait, not a wage-specific one.

Third, under the survey's density-implied variance, a parametric fit in the tradition of Engelberg et al. (2009), the group means are 28.6/16.4/11.3 against the midpoint measure's 20.8/11.2/11.3 on the same reports of the broader panel, a steeper gradient. The comparison qualifies Section 3: the middle-top equality there is a property of the midpoint measure, while the ordering and the bottom's excess survive under either measure.

C.5 Persistence of the realised volatility process

The belief-persistence gradient of Table 1 is not inherited from the realised volatility process: the squared changes of the realised series are if anything less persistent at the bottom than at the top. The check runs on the raw, unsmoothed cell series underlying the shock regressor and compares the bottom with the top, the pair whose belief persistence differs significantly.

The within-cell monthly AR(1) of the squared change is 0.29 at the bottom against 0.36 at the top, and 0.44 against 0.49 in the national series. Instrumenting the lag to remove finite-cell sampling noise drives the bottom to 0.06 against the top's 0.26, and the bottom's quarterly national series is mildly anti-persistent. The uninstrumented numbers overstate the bottom if anything: adjacent squared changes share a mechanical correlation through the same person-month, and that correlation is larger where cells are smaller, as they are at the bottom.

C.6 The realised process: higher-order moments and the stayer conditioning

The higher-order autocovariances of the wage change are economically second order, so the permanent-transitory model behind Table 2 is an adequate description of the realised process. The model implies $\text{Cov}(\Delta e_t, \Delta e_{t-k}) = 0$ for all $k \geq 2$, and at these sample sizes the estimated moments are statistically distinguishable from zero but small, between 2 and 23 percent of the lag-1 moment that identifies $\sigma^2_{\theta,g}$. The largest deviations sit at the quarterly lags for the top, +0.99 and +0.41 at lags three and six, the signature of quarterly bonus delivery (Section 7).

The stayer conditioning does not select the sample in a way that could understate the bottom's transitory risk. Months immediately preceding an employer change carry elevated volatility at the bottom, about 1.6 times other months, but they are one percent of the sample, so excluding them moves $\sigma^2_{\theta,g}$ by less than 0.04 (%)^2 in every group. Closing the bottom's prior gap through selective separation would instead require the conditioning to hide a fourfold understatement of its transitory risk, far beyond the observed elevation.

The sort itself is a choice the realised process is robust to. Sorting workers by their own annual earnings rather than by household income reassigns about half of them, since spousal income moves many workers across the household cut-offs, and it leaves the pattern of Table 2 intact: the transitory variance rises more steeply with income, at 2.28, 3.49, and 8.11 (%)^2 across own-earnings terciles, and the permanent variance stays roughly flat, at 12.44, 11.30, and 10.86. The realised gradient therefore runs opposite to the perceived one under either sort. The belief side cannot be re-sorted, because the SCE elicits household income alone, so the comparison across surveys keeps the household sort.

C.7 The same-hours counterpart of the realised process

The SCE question holds hours fixed, while the recorded SIPP wage carries them: earnings are divided by the average weekly hours reported for the month, premium pay and tips enter the numerator, and the reported schedule sets the divisor. The counterpart the conditioning calls for therefore restricts the wage changes to months in which the reported hours do not change. I recompute the moments of Table 2 keeping only consecutive within-spell changes across which the reported average weekly hours are unchanged, requiring both differences of each autocovariance pair to qualify and applying the full-sample trim bounds, so that the restriction and not the trim moves the estimates; within the restricted pairs the wage change is the change in earnings at a fixed divisor, which also frees it of the division bias the baseline carries. The restricted moments bound the object the question asks about from above rather than measure it, since reported hours are heaped, recorded changes undercount true ones, and hours worked still vary around an unchanged report. The restriction is mild, dropping 2.3, 1.8, and 1.4 percent of observations from the bottom up, so selection on the realised outcome has little room to operate, though the dropped share itself falls with income, one more form of the exposure gradient; what it removes is concentrated all the same, thirty-one percent of the total monthly variance at the bottom against sixteen at the top. A placebo that drops the same share of changes at random leaves the permanent variance within 0.13 (%)^2 of its baseline in every group, so it is the months removed, not the observations lost, that carry the change.

Group	Baseline $\sigma^2_{\theta,g}$	Baseline $\sigma^2_{\psi,g}$	Same reported hours $\sigma^2_{\theta,g}$	Same reported hours $\sigma^2_{\psi,g}$	Share kept
Bot	2.52	11.92	2.83	6.12	0.977
Mid	4.42	11.61	4.73	6.31	0.982
Top	5.83	11.23	6.07	7.13	0.986

Table 9: SIPP permanent-transitory identification with and without the same-hours restriction, monthly cadence, in $(\%)^2$. The baseline columns re-estimate the construction of Table 2 on the same code path, matching its moments to within 0.05; the same-hours columns keep only within-spell changes across which the reported average weekly hours are unchanged, with both differences of each autocovariance pair required to qualify. Share kept is the fraction of baseline observations surviving the restriction.

Table 9 sets the restricted moments against the baseline. The transitory variance is essentially unchanged and still rises with income, so the realised gradient runs opposite to the perceived one under either conditioning. The netting floors of Appendix D.3 move to 2.49, 3.89, and 3.90, the middle and top priors remain within one standard error of them, and the bottom's excess is 7.16 ($t = 4.1$) against 7.43 at the baseline; rescaling the labelled share to the restricted movement, which removes hours-event variance from its denominator, raises the shares to about 0.17, 0.23, and 0.43 and moves none of the three comparisons. The full-information attribution ratio $\lambda^*_{\text{Bot}}/\lambda^*_{\text{Top}}$ moves from 1.43 to 1.40, and the closed form (7) at the restricted moments predicts attribution ratios of 1.71 and 1.25 in place of 1.69 and 1.31, still within one standard error of the estimates; the persistence comparison of C.5 is equally insensitive, showing no bottom-top gap whether the wage is measured per reported hour or raw.

The permanent variance falls by roughly half at the bottom and the middle and by over a third at the top. A change in the reported schedule is rare but persistent, a move between full- and part-time work or a renegotiated contract rather than a noisy month, so the moments of Section 4 assign what it carries to the permanent component; the hours-linked permanent share, 5.8, 5.3, and 4.1 $(\%)^2$ from the bottom up, itself falls with income, the delivery gradient of Section 7 on one more margin. The twelve-month benchmark (1) falls to 79, 85, and 98 $(\%)^2$ and now rises with income where the reported variance falls, so the correction does not cancel from the cross-group comparison; it strengthens the contrast, and the reported variance sits about four to nine times below the tighter benchmark in every group, so the shortfall on which the constant-attribution evidence of Section 8 rests is preserved.

The removed variance also gives the bottom's prior excess a second possible source. A bottom worker unable to comply with the instruction, who folded part of the hours-event risk into her answer, would be reporting truthfully and would still sit above her netting floor, and folding in about a fifth of the removed variance would account for the excess in full. The moments cannot separate this partial inclusion from unresolved reading of the pay she does condition on, while the inflation-density widening of Appendix C.4, which no hours channel reaches, indicates that part of the excess is generic reporting rather than either. Little turns on the allocation for behaviour, since each reading enters the transitory prior and not the permanent belief, but the excess is best described as reported risk beyond the object the question asks about, with reading noise the component the mechanism predicts rather than the whole the data establish.

The baseline moments remain those of Table 2, because the budget of Section 9 must carry the earning-power risk the household actually absorbs, and a persistent cut in scheduled hours is exactly such risk; the question conditions it out of the elicited object, not out of her income. The cost of the choice is bounded in closed form: at the restricted moments and the measured shares the resting discounts $V_u/\bar{S}_m$ deepen to 0.94, 0.89, and 0.77 while the twelve-month permanent gaps shrink from −5.0, −11.9, and −25.3 to −4.3, −8.1, and −19.3 $(\%)^2$, so the wedge of Section 10 keeps its shape and its concentration at the top at magnitudes smaller by between a seventh and a third.

C.8 The benchmark at long horizons

The twelve-month benchmark of Section 4 extrapolates monthly moments under the random walk rather than measuring the twelve-month change, so this subsection measures it directly on the same panel. For each horizon h I take within-spell pairs h months apart, drop the pair whose later endpoint is January as the monthly construction does, and trim the squared change at the same one and ninety-nine percent. At $h = 1$ the exercise returns the moments of Table 2 by construction, 16.9, 20.4, and 22.9 $(\%)^2$. At $h = 12$ it returns 2724, 2610, and 2774, about eighteen times the extrapolated benchmark.

The gap is a statement about outliers rather than about the wage process. A job stayer's monthly change is a spike at zero with heavy tails, since most months carry no change in the reported rate, so the mean-based moment is governed by the tail and the common trim removes about eighty-five percent of it at one month against under half at twelve. Measuring the same profile with a scale that ignores the tails, the interquartile range of the change in Gaussian variance units, moves the comparison the other way: it puts the monthly variance near 0.6, far below the trimmed moment, and the twelve-month variance near 340, twice the benchmark rather than eighteen times it. One measure is governed by the tail and the other by the spike, and the data do not discipline the choice between them.

The level of the benchmark is therefore not identified without a model of the reporting outliers, which is why Section 4 reads it as an upper bound and rests the argument on the contrast across groups rather than on the level. What the profile does settle is the direction: the directly measured twelve-month variance lies above the extrapolation under either scale, so the shortfall of reported risk against realised risk is not an artefact of the random-walk extrapolation.

D Identification: the filter, the labelled share, and the scorecard

This appendix gives the filter behind the gain (D.1), the measurement of the labelled share and its robustness (D.2), the scorecard setting the measured mechanism against the estimated belief moments (D.3), the filter's diagnostic on the estimated gains (D.4), and the hours evidence behind the reading-noise gradient (D.5).

D.1 The filter

Once the attribution is rescaled to the agent's own signal, the perceived transitory prior drops out of the filter's signal-to-noise ratio, so the gain depends on the labelled share and not on reading noise.

The agent's monthly signal is her own squared netted reading, $x_t = (\Delta\tilde{y}_t)^2$ with $\tilde{y}$ from (5). It has mean $\sigma^2_{\psi,g} + V_{x,g}$ and, under Gaussianity, variance $2(\sigma^2_{\psi,g} + V_{x,g})^2$, where $V_{x,g} = 2\sigma^2_{\theta,\text{perc},g}$ is twice the perceived transitory prior. The perceived permanent variance follows a random walk with innovation variance q (the drift), the filter's one free parameter, fitted to the estimated gains in D.4. The observation is the attributed signal

$$z_t \;=\; \tilde{\lambda}_g\, x_t \;=\; \hat{\sigma}^2_{\psi,t} + \varepsilon_t,$$

a unit-loading reading of the state, where $\tilde{\lambda}_g = \sigma^2_{\psi,g} V_{u,g}/[\bar{S}_{m,g}(\sigma^2_{\psi,g} + V_{x,g})]$ is the full-change attribution (7) rescaled to the signal's mean and $V_{u,g} = \sigma^2_{\psi,g} + 2(1-\rho_g)\sigma^2_{\theta,g}$ is the unlabelled residual variance of the main text. The rescaling makes the observation unbiased at rest, since $\tilde{\lambda}_g\,\mathbb{E}[x_t] = \sigma^2_{\psi,g}\, V_{u,g}/\bar{S}_{m,g}$ is the permanent belief (8). The variance of a squared signal moves with the state, so I evaluate it at the resting point and read the filter as moment-matched rather than exact. Its steady-state gain is

$$\mu_g \;=\; \frac{\bar{P}_g}{\bar{P}_g + R_g}, \qquad \bar{P}_g \;=\; \tfrac{1}{2}\big(q + \sqrt{q^2 + 4qR_g}\big),$$

with $\bar{P}_g$ the positive root of the Riccati equation, and the observation noise is

$$R_g \;=\; \mathrm{Var}(z_t) \;=\; \big(\tilde{\lambda}_g\big)^2\, 2(\sigma^2_{\psi,g} + V_{x,g})^2 \;=\; 2(\sigma^2_{\psi,g})^2\,(V_{u,g}/\bar{S}_{m,g})^2.$$

The prior enters the attribution and the signal variance by offsetting factors and cancels from R_g, so the gain depends on the labelled share, through $V_{u,g}$, and not on reading noise. The model carries the gain as the estimated moment, and the filter serves only as the D.4 diagnostic on the gains' ordering.

D.2 Measuring the labelled share

The labelled share of within-spell pay movement rises with income, from 0.121 at the bottom through 0.178 in the middle to 0.358 at the top, and the gradient survives every specification in Table 10. I measure it on the stayer panel of Section 2: 2,250,815 person-months across 103,134 persons and 127,184 spells.

For a worker i in job spell s, let y_{it} be total earnings in month t and ℓ_{it} labelled pay, bonuses and commissions in the main specification, with spell means $\bar{y}_s$ and $\bar{\ell}_s$. I demean each within the spell and scale by spell-mean earnings,

$$d^{\text{tot}}_{it} = \frac{y_{it} - \bar{y}_s}{\bar{y}_s}, \qquad d^{\text{lab}}_{it} = \frac{\ell_{it} - \bar{\ell}_s}{\bar{y}_s},$$

and take the person-weighted coefficient from regressing the labelled deviation on the total through the origin,

$$\rho_g = \frac{\sum_{(i,t)\in g} w_{it}\, d^{\text{lab}}_{it}\, d^{\text{tot}}_{it}}{\sum_{(i,t)\in g} w_{it}\, \big(d^{\text{tot}}_{it}\big)^2} = \frac{\mathrm{Cov}_w\big(d^{\text{lab}}, d^{\text{tot}}\big)}{\mathrm{Var}_w\big(d^{\text{tot}}\big)},$$

where w_{it} are the SIPP person weights and the sum runs over stayer person-months in group g. Seam months are dropped as in Section 2.

Writing $d^{\text{tot}} = d^{\text{lab}} + d^{\text{unlab}}$ gives $\rho_g = [\text{Var}_w(d^{\text{lab}}) + \text{Cov}_w(d^{\text{lab}}, d^{\text{unlab}})]/\text{Var}_w(d^{\text{tot}})$, which reduces to the pure variance share when the two parts are uncorrelated, the alternative row of Table 10 and close to the main one throughout. The denominator carries permanent drift and any tenure trend, so ρ_g is a conservative proxy for the labelled share of $\sigma^2_{\theta,g}$ alone.

Specification	Bot	Mid	Top
Main: within-spell demeaned, labelled = bonus + commission	0.121	0.178	0.358
Variance share instead of regression share	0.146	0.203	0.373
Within-spell detrended	0.176	0.256	0.488
Imputed component months dropped	0.042	0.085	0.314
Overtime counted as labelled	0.134	0.197	0.370
Annual spell-year cells (different frequency)	0.050	0.060	0.075

Table 10: The measured labelled share ρ_g across specifications, by income group, SIPP stayer panel 2013–2023, person-weighted. The main row is the central specification, and the model pins its share within the band these rows span (Section 9).

The remaining rows set the measurement band. The detrended row removes a within-spell linear trend that demeaning leaves in the unlabelled residual and bounds the share from above. The imputation row bounds it from below: imputation that spreads labelled dollars across months can manufacture labelled variance, so the row drops every imputed person-month, lowering the shares but leaving the gradient intact; it is the band's lower edge, since it also discards genuine labelled months. The annual row does not enter the band: at spell-year cells the monthly variation the filter reads has averaged out, so it measures the labelled fraction of year-to-year risk.

Receipt shares in the same records confirm the direction of the delivery gradient. Bonus receipt rises from 2.8 to 4.8 to 9.7 percent across the groups in 2018 (2.8/7.1/12.1 in 2021), receipt of any labelled component from 7.9 to 16.5 percent between bottom and top, and receipt of tips falls from 2.4 to 0.9. Receipt speaks to direction, not the level of ρ, and places the middle below the top on every labelled component.

A final check: the netting floor $(1 - \rho_g)\sigma^2_{\theta,g}$ that D.3 scores the priors against multiplies a level-based share by the trimmed difference moment of Section 4. Measured directly as the lag-1 autocovariance of the unlabelled pay change, the floor moves by about a tenth or less under the trim in every group, while the total difference moment moves by factors of seven to fifty, because the trimmed tail is almost entirely labelled bonus and commission months. The labelled share implied by the difference moments, 0.08/0.10/0.48, lies inside the band at the bottom and the middle and at its detrended upper edge at the top, so the top's floor comparison supports the band rather than the point.

D.3 The scorecard

With the labelled share measured, the estimated belief moments of Section 6 become out-of-sample predictions, scored by three comparisons. The floors isolate rather than predict a fourth number, the bottom's residual.

The first comparison sets the estimated priors against the netting floors $(1-\rho_g)\sigma^2_{\theta,g} = 2.21/3.63/3.74$. The middle and top priors sit on them: the residuals $\sigma^2_{\theta,\text{perc},g} - (1-\rho_g)\sigma^2_{\theta,g} = \hat{\omega}^2_g$ of (6) are -0.68 and $+0.38$, with standard errors 0.94 and 0.82, both within one standard error of zero.

The middle's estimate sits 0.7 standard errors below its floor, an insignificant violation of the inequality $\omega^2 \geq 0$; the model carries the censored value. The bottom's residual, +7.43 (standard error 1.76, $t = 4.2$), is the identification of the reading noise, concentrated where hours volatility is highest (D.5). The comparisons repeat under the same-hours moments of Appendix C.7, with residuals of -0.93 (0.94) at the middle, +0.22 (0.82) at the top, and +7.16 (1.76) at the bottom.

Reporting error in either survey runs against the bottom's residual rather than for it. Classical error in SIPP earnings loads one-for-one into measured $\sigma^2_{\theta,g}$ and hence into the floors, so at an error share s of transitory variance each residual rises by $s\,\sigma^2_{\theta,g}$: the middle's stays within one standard error of zero for s up to 0.37 and insignificant up to 0.57, the top's crosses one standard error at $s = 0.08$ and significance at 0.21, and the bottom's excess only grows.

On the SCE side, a common understatement of uncertainty would deflate every prior and understate the excess further, while an overstatement large enough to manufacture it, a factor near four, would push the middle and top priors far below floors they instead sit on. The group-specific overstatement that remains is the numeracy channel, conditioned on in Appendix C.4.

The second comparison runs on cross-group attribution ratios rather than the level, because the cell-mean shock regressor carries a worker's own shock only in part, so the slope recovers her response up to a loading the merge itself does not pin down. The loading can be measured in the SIPP: regressing each worker's own trimmed $(\Delta e_{i,t})^2$ on the cell mean of the other workers in her cell, constructed exactly as the regressor is, gives 0.43, 0.56, and 0.58 from the bottom up.[13] The loadings are similar though not identical, and the difference runs in the conservative direction: correcting the ratios by them raises bottom-to-top from 3.04 to about 4.1 and moves middle-to-top from 1.76 to 1.83, so treating the loading as common understates the gradient the mechanism is asked to explain rather than manufacturing it.

At the measured shares the closed form (7) predicts bottom-to-top and middle-to-top attribution ratios of 1.69 and 1.31 (in logs 0.52 and 0.27), against estimated ratios of 3.04 and 1.76 (in logs 1.11 and 0.57, with log standard errors 0.63 and 0.59). The gaps are $z = +0.9$ and $+0.5$, within one standard error, and the predictions are about half of the estimated point gradient. Under the measured-loading correction the estimated gradient widens and the mechanism's share falls to about a third at the bottom; I keep the uncorrected comparison as the baseline, since the loading is itself an approximation, and read the corrected one as the outer edge. The moment has little power against full information: the FIRE gradient $\lambda^*_{\text{Bot}}/\lambda^*_{\text{Top}} = 1.43$ (log 0.36) sits $z = 1.2$ from the

[13]The projection is the least-squares loading, an approximation to the factor the system estimator recovers through lagged instruments. Its common variation is in large part the seasonal raise cycle, which under the rule is genuine updating variation, the realised squared changes the agent responds to whether or not their season is predictable; stripping it, by deseasonalising the cell series by month-of-year or by pushing the instruments beyond the smoothing window, weakens all three estimated responses and degrades the estimator's moment conditions, so the news response is identified largely off the seasonal arrival of wage news.

estimate, so the ratios discipline the mechanism's direction and rough size, not its existence.

The third comparison is the gain diagnostic of D.4, which predicts the gains' pinned contrast, the bottom slowest, and underpredicts their spread, on the moments the cadence comparison of Section 3 marks as the least reliable. The tally is two moment sets consistent within one standard error, one ordering the filter gets right while underpredicting its spread, and one significant residual the mechanism names rather than predicts.

D.4 The gain diagnostic

The filter of D.1, run at the measured shares, predicts the ordering of the estimated gains but underpredicts their spread. The observation noise is $R_g = 2(\sigma^2_{\psi,g})^2(V_{u,g}/\bar{S}_{m,g})^2 = 262.7/229.3/168.3$, which falls with the labelled share, so any common drift q implies a gain that rises with income. At the common drift that best matches the three estimated gains, weighted by their precisions, $q^* = 646.5$, the predicted gains are 0.763/0.783/0.823 against the estimated 0.629/0.910/0.950. The Gaussian factor of two in R_g is not load-bearing: the kurtosis of the trimmed residualised wage change is heavy but nearly common across groups (56.5/54.1/52.3), so the empirical observation noise rescales the three groups together, and the refitted drift absorbs the common factor, leaving the predicted ordering and the spread shortfall unchanged.

The diagnostic also separates the constant attribution from the Bayesian rescaling of Section 8. The rescaler's attributed signal has variance $2(\sigma^2_{\psi,g})^2 = 282.7/269.1/251.8$, so any common drift implies a nearly flat gain profile, at 0.79/0.79/0.80 under the same fit, against the estimated rise from 0.63 to 0.95. The noise of the constant-attribution filter falls with the labelled share instead, and it predicts the rising order.

Experience-based learning grades updating by age (Malmendier and Nagel, 2011), so an income gradient in the gain could be an age gradient in disguise. Within the 40-plus age band, however, the estimated gain rises from 0.57 to 0.84 to 0.98 across the income groups, a bottom-to-top gap of over four standard errors. The under-40 band retains too few bottom-group respondents (117) to estimate the gain, so the check rests on the older band alone.

D.5 Hours by income group

Within-person hours volatility $\mathbb{E}[(\Delta \log h)^2]$ falls monotonically with income, 4.10/3.06/2.36 in $(\%)^2$ (Figure 10), the direction the reading-noise residual of D.3 requires. The reported hours variable is heaped, so nearly all of this volatility comes from the rare schedule changes that Appendix C.7 nets out of the realised process; the gradient therefore measures a worker's exposure to schedule events, and it bears on the reading problem as a proxy, since the within-schedule movement she must read through is exactly what a heaped report cannot record. Transitory shocks delivered through hours are also of the unlabelled kind, so variable hours lower ρ_g as well. The earnings stream itself is lumpiest at the bottom: the kurtosis of the trimmed raw log earnings change falls from 46.8 through 24.1 to 15.5 across the groups, where the residualised wage change of D.4 is nearly common, so the flow she must read is more leptokurtic as well as more volatile. The check is directional only: a proportional scaling of the residual in hours volatility, calibrated at the bottom (7.43/4.10), would predict a middle residual near 5.5 against an estimated −0.68

(standard error 0.94), so no intensity mapping from observables to ω^2 is estimable.

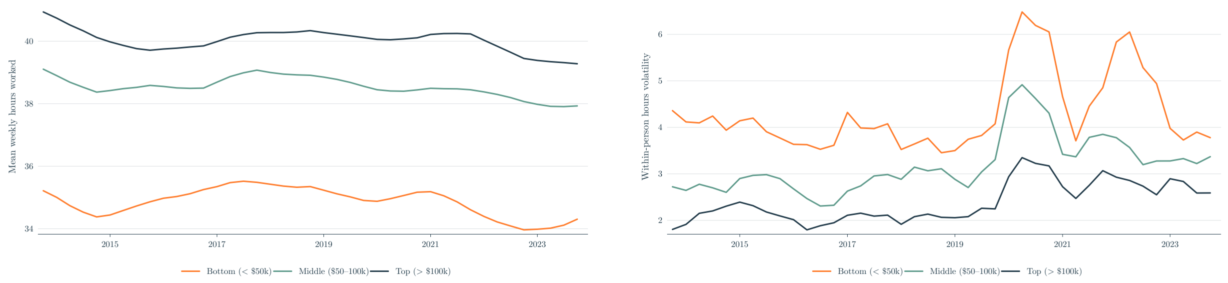

Figure 10: Mean weekly hours (left) and within-person hours volatility $\mathbb{E}[(\Delta \log h)^2 \mid g, t]$ in $(\%)^2$ (right) by income group. SIPP matched-stayer panel, quarterly with 4-quarter trailing MA.

E Calibrating the belief profile

This appendix builds the continuous belief profile the model consumes and defines the upper-bound variant that Section 10 reports.

E.1 From three groups to a continuum

The terciles are the finest income split the SCE powers, so the profiles of Figure 5 interpolate the three groups rather than estimate finer bins. Re-estimating on the five native SCE brackets breaks the belief block where it is split: the lowest bracket (under $20k, 102 respondents) has no estimable cell, the attribution turns negative in the lower-middle, the gain exceeds one in the top two, and only the middle, the unchanged tercile, survives.

Interpolation. The share ρ and the gain μ interpolate in levels and the prior $\sigma^2_{\theta,\text{perc}}$ in logs, monotone cubic through the three anchors and flat outside them. End tangents are zero, so each profile joins its flat extension without a kink; interior tangents are bounded so every segment stays monotone. The prior's floor censoring at $(1 - \rho(y))\,\sigma^2_\theta(y)$ binds only at the middle anchor itself.

Table 11 evaluates the mapping at representative incomes. The attribution, the discount, and the permanent belief follow from the closed forms: the λ column is the estimating form (7), and $\tilde{\lambda}$ is the implemented form on the agent's own netted reading.

Monthly income	Carried ρ	Carried μ	$\sigma^2_{\theta,\text{perc}}$	Attribution λ	$\tilde{\lambda}$	Discount $V_u/\bar{S}_m$	Perceived perm. var
\$1,500	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$2,415 (Bot)	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$3,500	0.134	0.718	6.72	0.610	0.445	0.952	11.21
\$6,014 (Mid)	0.197	0.910	3.55	0.519	0.568	0.915	10.61
\$9,000	0.297	0.945	3.82	0.453	0.515	0.860	9.81
\$13,604 (Top)	0.369	0.950	4.12	0.398	0.468	0.812	9.11
\$20,000	0.369	0.950	4.12	0.398	0.468	0.812	9.11

Table 11: The income-to-belief mapping of Figure 5 evaluated at representative incomes. Rows away from the three group anchors (Bot, Mid, Top) are interpolated, not separately measured. In $(\%)^2$ except the dimensionless shares.

E.2 The upper-bound variant

The belief-anchored variant takes the estimated attribution responses at face value rather than reading them through the measured delivery of pay. It anchors the bottom's attribution at its full-information share $\lambda^* = \sigma^2_\psi/\bar{S}_m = 0.702$ and lets the estimated attribution ratios supply the middle and the top, giving $\lambda = 0.702/0.407/0.231$; the resting permanent belief $\lambda\,\bar{S}_m$ is then 11.89/8.32/5.29, discounts of 1.00/0.72/0.47 on the truth. The prior is carried as in the baseline. The variant's wedge is 8.5, 19.5, and 32.4 percent of wealth and +1.2, +3.1, and +6.7 points on the marginal propensity, the same shape as the measured calibration at roughly twice the size. The gap between the two calibrations is the part of the wedge that rides on netting beyond the measured components, or on sampling noise in ratios this wide.

F Steady-state model output

Table 12 gives the perceived variance under the measured beliefs against FIRE, by income group.

Group	$\hat{\sigma}^2_\psi$	$\sigma^2_{\theta,\text{perc}}$	Perceived $\bar{\sigma}^2_{\text{perc}}$	FIRE	Ratio
Bot	11.3	8.3	153	148	1.03
Mid	10.8	5.1	139	148	0.94
Top	9.7	3.9	124	147	0.84

Table 12: Stationary perceived variance against FIRE at the 12-month horizon, in $(\%)^2$: simulated cross-sectional means by tercile under mobility. $\hat{\sigma}^2_\psi$ is the mean permanent belief and $\sigma^2_{\theta,\text{perc}}$ the mean transitory prior, with realised counterparts 11.8/11.6/11.4 and 2.9/4.0/5.3; $\bar{\sigma}^2_{\text{perc}} = 12\hat{\sigma}^2_\psi + 2\sigma^2_{\theta,\text{perc}}$ and FIRE $= 12\sigma^2_\psi + 2\sigma^2_\theta$. Entries are computed from unrounded inputs, so displayed sums and ratios can differ in the last digit.

Table 13 gives the numbers behind Figure 7.

Group	Mean wealth (months) FIRE	Belief	Δ	12-month MPC FIRE	Belief	Δ
Bot	2.28	2.17	−5.1%	0.470	0.477	+0.7pp
Mid	1.71	1.57	−8.3%	0.411	0.423	+1.3pp
Top	1.50	1.29	−13.8%	0.377	0.400	+2.4pp
Workers only, re-terciled among workers						
Bot	2.20	2.07	−5.8%	0.334	0.343	+0.9pp
Mid	1.54	1.39	−9.7%	0.359	0.374	+1.5pp
Top	1.47	1.25	−15.1%	0.348	0.375	+2.7pp

Table 13: Belief regime versus FIRE by income tercile. Wealth is in months of permanent income, the 12-month MPC is the annualisation $1 - (1 - m)^{12}$ of the one-month response m to a transitory windfall, and the regimes are paired by common random numbers. Partial equilibrium at $r = 2$ percent with common $\beta = 0.917$, which targets mean liquid wealth of about 1.8 months. The bottom tercile is 54 percent retirees, against 22 and 12 percent above; the workers-only panel re-terciles among workers. Its MPC levels sit below the full-population levels and are not monotone, because removing retirees removes short horizons and high propensities, while the belief wedge itself sharpens.

Table 14 gives the general-equilibrium counterpart of Section 10.

Regime	r (% ann.)	K	H2M share
FIRE	3.00	3.65	0.338
Belief	3.21	3.45	0.342

Table 14: General-equilibrium FIRE versus belief, common $\beta = 0.917$, one-asset closure, with Z pinned from FIRE at $r^* = 3\%$ and held fixed across regimes. H2M is the hand-to-mouth share.